

Chapter IX

High Pressure Piping

(24) K300 GENERAL STATEMENTS

(a) Chapter IX provides alternative rules for design and construction of piping designated by the owner as being in High Pressure Fluid Service. See para. 300(b)(1) and Appendix M.

(1) Use of this Chapter is permitted only when the owner designates the piping as being in High Pressure Fluid Service. When piping is so designated, the requirements of this Chapter apply in their entirety.

(2) There are no pressure limitations for the application of these rules. See Appendix F, para. FK300.

(b) Responsibilities. In addition to the responsibilities stated in para. 300(b)

(1) for piping designated as being in High Pressure Fluid Service, the owner shall provide all system operations information necessary for the designer to perform the analyses and testing required by this Chapter

(2) the designer shall provide the owner with a written report that includes the following:

(-a) list of all technical references used
 (-b) list of all computer software and versions used
 (-c) list of all assumptions used in the design of the piping system

(-d) list of design pressures and temperatures and the portions of piping to which applied throughout each piping system

(-e) ambient and thermal expansion and contraction effects considered in the flexibility analysis

(-f) specific conditions (e.g., two-phase flow, surges, slugs, vibration, relief loads, acoustic resonance) addressed in the dynamic analysis

(-g) list and description of unlisted materials and components

(-h) list and description of all pressure relief devices

(-i) design calculation reports

(-j) certification that the design has been performed in accordance with this Chapter

(c) The organization, content, and, wherever possible, paragraph designations of this Chapter correspond to those of the base Code (Chapters I through VI). The prefix K is used.

(d) Provisions and requirements of the base Code apply only as stated in this Chapter.

(e) Paragraphs 300(a) and 300(c) through 300(f) apply, but in para. 300(c)(3), refer to para. K302 instead of para. 302.

K300.1 Scope

K300.1.1 Content and Coverage. Paragraph 300.1.1 applies with the exceptions stated in paras. K300.1.3 and K300.1.6.

K300.1.2 Packaged Equipment Piping. Interconnecting piping as described in para. 300.1.2 shall conform to the requirements of this Chapter.

K300.1.3 Exclusions. In addition to the exclusions stated in para. 300.1.3, this Chapter excludes nonmetallic piping and piping lined with nonmetals.

K300.1.4 Units of Measure. Paragraph 300.1.4 applies.

K300.1.5 Rounding. Paragraph 300.1.5 applies.

K300.1.6 Category M Fluid Service. This Chapter makes no provision for piping in Category M Fluid Service. If such piping is required by the owner, the engineering design shall be developed as provided in para. 300(c)(5).

K300.2 Definitions

Paragraph 300.2 applies except for terms relating only to nonmetals and severe cyclic conditions.

The term “allowable stress” is used in lieu of basic allowable stress.

The term “safeguarding” and other terms characterizing hazardous fluid services are not used in this Chapter but should be taken into account in design.

K300.3 Nomenclature

Paragraph 300.3 applies.

K300.4 Status of Appendices

Paragraph 300.4 and Table 300.4-1 apply, except for Appendices A, B, H, L, R, S, V, and X.

PART 1 CONDITIONS AND CRITERIA

K301 DESIGN CONDITIONS

Paragraph 301 applies with the exceptions of paras. 301.2, 301.3, 301.4.2, 301.5, and 301.7.3.

K301.2 Design Pressure

K301.2.1 General. Paragraph 301.2.1 applies, except that references to para. 302.2.4 are not applicable and refer to para. K304 instead of para. 304.

K301.2.2 Required Pressure Containment or Relief. Paragraphs 301.2.2(a) and 301.2.2(b) apply, but refer to para. K322.6.3 instead of para. 322.6.3. Paragraph 301.2.2(c) is not applicable.

K301.3 Design Temperature

Paragraph 301.3 applies with the exceptions of paras. 301.3.1 and 301.3.2 and the following exceptions in the text:

- (a) Refer to para. K301.2 instead of para. 301.2.
- (b) Refer to para. K301.3.2 instead of para. 301.3.2.

K301.3.1 Design Minimum Temperature. Paragraph 301.3.1 applies, but refer to para. K323.2.2 instead of para. 323.2.2.

- (24) **K301.3.2 Uninsulated Piping Components.** The component design temperature shall be the fluid temperature unless a warmer temperature will result from solar radiation or other external heat sources.

K301.4 Ambient Effects

K301.4.2 Fluid Expansion Effects. Paragraph 301.4.2 applies, except that reference to para. 322.6.3(b)(2) is not applicable.

K301.5 Dynamic Effects

Paragraph 301.5 applies with the exception of para. 301.5.4.

K301.5.4 Vibration. Suitable dynamic analysis shall be made where necessary, to avoid or minimize conditions that lead to detrimental vibration, pulsation, or resonance effects in the piping.

K301.7 Thermal Expansion and Contraction Effects

K301.7.3 Loads Due to Differences in Expansion Characteristics. Paragraph 301.7.3 applies, except that reference to metallic-nonmetallic piping is not applicable.

K302 DESIGN CRITERIA

K302.1 General

In para. K302, pressure-temperature ratings, stress criteria, design allowances, and minimum design values are stated, and permissible variations of these factors as applied to design of high pressure piping systems are formulated.

The designer shall be satisfied as to the adequacy of the design, and of materials and their manufacture, considering at least the following:

- (a) tensile, compressive, flexural, and shear strength at design temperature
- (b) fatigue strength
- (c) design stress and its basis
- (d) ductility and toughness
- (e) possible deterioration of mechanical properties in service
- (f) thermal properties
- (g) temperature limits
- (h) resistance to corrosion and erosion
- (i) fabrication methods
- (j) examination and testing methods
- (k) hydrostatic test conditions
- (l) bore imperfections

K302.2 Pressure-Temperature Design Criteria

K302.2.1 Listed Components Having Established Ratings. Pressure-temperature ratings for certain piping components have been established and are contained in some of the standards in Table K326.1.1-1. Unless limited elsewhere in this Chapter, those ratings are acceptable for design pressures and temperatures under this Chapter. With the owner's approval, the rules and limits of this Chapter may be used to extend the pressure-temperature ratings of a component beyond the ratings of the listed standard, but not beyond the limits stated in para. K323.2.

K302.2.2 Listed Components Not Having Specific Ratings

(a) Piping components for which design stresses have been developed in accordance with para. K302.3, but that do not have specific pressure-temperature ratings, shall be rated by rules for pressure design in para. K304, within the range of temperatures for which stresses are shown in Table K-1 (Table K-1C), modified as applicable by other rules of this Chapter.

(b) Piping components that do not have allowable stresses or pressure-temperature ratings shall be qualified for pressure design as required by para. K304.7.2.

K302.2.3 Unlisted Components. Piping components not listed in Table K326.1.1-1 may be used subject to all of the following requirements:

- (a) The material shall comply with para. K323.

(b) The designer shall be satisfied that the design is suitable for the intended service.

(c) Pressure-temperature ratings shall be established in accordance with the rules in [para. K304](#).

(d) Fatigue analysis shall be performed as required by [para. K304.8](#).

K302.2.4 Allowance for Pressure and Temperature Variations. Variations in pressure, temperature, or both above the design conditions, except during pressure-relieving events (see [para. K322.6.3](#)), are not permitted for any piping system. The design pressure and design temperature resulting in the most severe coincident pressure and temperature shall determine the design conditions. See [paras. K301.2](#) and [K301.3](#).

K302.2.5 Ratings at Junction of Different Services. [Paragraph 302.2.5](#) applies.

K302.3 Allowable Stresses and Other Design Limits

K302.3.1 General. The allowable stresses defined below shall be used in design calculations unless modified by other provisions of this Chapter.

(a) *Tension.* Allowable stresses in tension for use in design in accordance with this Chapter are listed in [Table K-1 \(Table K-1C\)](#), except that maximum allowable stress values and design stress intensity values for bolting, respectively, are listed in ASME BPVC, Section II, Part D, Tables 3 and 4.

The tabulated stress values in [Table K-1 \(Table K-1C\)](#) are grouped by materials and product form and are for stated temperatures up to the limit provided for the materials in [para. K323.2.1](#). Straight line interpolation between temperatures to determine the allowable stress for a specific design temperature is permissible. Extrapolation is not permitted.

(b) *Shear and Bearing.* Allowable stress in shear shall be 0.57 times S_{yt} . Allowable stress in bearing shall be 1.0 times S_{yt} . See [para. K302.3.2\(b\)](#) for definition of S_{yt} .

(c) *Compression.* Allowable stress in compression shall be no greater than the allowable stress in tension tabulated in [Table K-1 \(Table K-1C\)](#). Consideration shall be given to structural stability.

(d) *Fatigue.* Allowable values of stress amplitude, which are provided as a function of design life in ASME BPVC, Section VIII, Division 3, Article KD-3, may be used in fatigue analysis in accordance with [para. K304.8](#).

(24) **K302.3.2 Bases for Allowable Stresses.** The bases for establishing allowable stress values for materials in this Chapter are as follows:

(a) *Bolting Materials.* The criteria of ASME BPVC, Section II, Part D, Appendix 2, 2-120 or 2-130, or Section VIII, Division 3, Article KD-6, KD-620, as applicable, apply.

(b) *Other Materials.* Allowable stress values at temperature for materials other than bolting materials shall not exceed the lower of the values calculated from [eqs. \(31a\)](#) and [\(31b\)](#). S_{ut} and S_{yt} shall be determined in accordance with [eqs. \(31d\)](#) and [\(31c\)](#), except that S_{ut} and S_{yt} shall not exceed S_T and S_Y , respectively.

$$S = 0.924K_{ut}S_{yt} \quad (31a)$$

$$S = \frac{1}{3}(S_{ut} + S_{yt}) \quad (31b)$$

$$S_{yt} = S_Y R_Y \quad (31c)$$

$$S_{ut} = 1.1S_T R_T \quad (31d)$$

where

K_{ut} = hydrostatic test pressure upper limit factor

= 0.95 for $S_Y/S_T \leq 0.7$

= $1.244 - 0.42(S_Y/S_T)$ for $0.7 < S_Y/S_T \leq 0.9$

= 0.866 for $S_Y/S_T > 0.9$

R_T = ratio of the average temperature dependent trend curve value of tensile strength to the room temperature tensile strength

R_Y = ratio of the average temperature dependent trend curve value of yield strength to the room temperature yield strength

S = allowable stress

S_T = specified minimum tensile strength at room temperature

S_{ut} = tensile strength at temperature

S_Y = specified minimum yield strength at room temperature

S_{yt} = yield strength at temperature

(c) *Unlisted Materials.* For a material that conforms to [para. K323.1.2](#), allowable stress values at temperature shall be calculated in accordance with (b).

(1) Except as provided in (2), S_{ut} and S_{yt} shall be determined in accordance with [eqs. \(31d\)](#) and [\(31c\)](#), respectively.

(2) If the tensile and yield strengths at temperature for an unlisted material are contained in ASME BPVC, Section II, Part D, Tables U-1 and Y-1, respectively, those tensile and yield strengths at temperature values may be used directly in the determination of allowable stress.

(d) *Cyclic Stresses.* Allowable values of alternating stress shall be in accordance with ASME BPVC, Section VIII, Division 3, Article KD-3.

K302.3.3 Castings.¹ Cast piping components shall conform to all of the following requirements: (24)

¹See Notes to [Tables 302.3.3-1](#) and [302.3.3-2](#) for titles of standards referenced herein.

Table K302.3.3-1
Acceptable Severity Levels for Steel Castings

Thickness Examined, mm (in.)	Applicable Standards	Acceptable Severity Level	Acceptable Discontinuity Categories
$\bar{T} \leq 51$ (2.0)	ASTM E446	1	A, B, C
$51 < \bar{T} \leq 114$ (4.5)	ASTM E186	1	A, B, C
$114 < \bar{T} \leq 305$ (12.0)	ASTM E280	1	A, B, C

(a) All surfaces shall have a roughness average, R_a , not greater than $6.3 \mu\text{m}$ R_a (250 $\mu\text{in.}$ R_a); see ASME B46.1 for a definition of R_a .

(b) All nonferromagnetic surfaces shall be examined using the liquid penetrant method in accordance with ASTM E165/E165M, with acceptability judged in accordance with MSS SP-93, Table 1. All ferromagnetic surfaces shall be examined using either the liquid penetrant method or the magnetic particle method, in accordance with ASTM E165/E165M or ASTM E709, respectively. Acceptability of imperfections, including those in weld repairs, shall be judged in accordance with MSS SP-93, Table 1 or MSS SP-53, Table 1, respectively.

(c) Each casting shall be fully examined either ultrasonically in accordance with ASTM E114 or radiographically in accordance with ASTM E94/E94M. Cracks and hot tears (Category D and E discontinuities in accordance with the standards listed in Table K302.3.3-1) and imperfections whose depths exceed 3% of nominal wall thickness are not permitted. Acceptable severity levels for radiographic examination of castings shall be in accordance with Table K302.3.3-1.

K302.3.4 Weld Joint Quality Factor. Piping components containing welds shall have a weld joint quality factor $E_j = 1.00$ (see Table 302.3.4-1), except that the acceptance criteria for these welds shall be in accordance with para. K341.3.2. Spiral (helical seam) welds are not permitted.

K302.3.5 Limits of Calculated Stresses Due to Sustained Loads and Displacement Strains

(a) *Internal Pressure Stresses.* Stresses due to internal pressure shall be considered safe when the wall thickness of the piping component, including any reinforcement, meets the requirements of para. K304.

(b) *External Pressure Stresses.* Stresses due to external pressure shall be considered safe when the wall thickness of the piping component, and its means of stiffening, meet the requirements of para. K304.

(c) *Stresses Due to Sustained Loads, S_L .* The stresses due to sustained loads, S_L , in any component in a piping system (see para. K320) shall not exceed S_h , where S_h is the allowable stress provided in Table K-1 (Table K-1C) at the metal temperature for the operating condition being considered.

(d) *Allowable Displacement Stress Range, S_A .* The computed displacement stress range, S_E , in a piping system (see para. 319.4.4) shall not exceed the allowable displacement stress range, S_A (see para. 319.2.3), calculated by

$$S_A = 1.25S_c + 0.25S_h \quad (32)$$

where

S_c = allowable stress from Table K-1 (Table K-1C) at minimum metal temperature expected during the displacement cycle under analysis

S_h = allowable stress from Table K-1 (Table K-1C) at maximum metal temperature expected during the displacement cycle under analysis

K302.3.6 Limits of Calculated Stresses Due to Occasional Loads

(a) *Operation.* Stresses due to occasional loads may be calculated using the equations for stress due to sustained loads in para. K320.2. The sum of the stresses due to sustained loads, such as pressure and weight, S_L , and of the stresses produced by occasional loads, such as wind and earthquake, shall not exceed 1.2 times the allowable stress provided in Table K-1 (Table K-1C). Wind and earthquake forces need not be considered as acting concurrently.

(b) *Test.* Stresses due to test conditions are not subject to the limitations in para. K302.3. It is not necessary to consider other occasional loads, such as wind and earthquake, as occurring concurrently with test loads.

K302.4 Allowances

In determining the minimum required thickness of a piping component, allowances shall be included for corrosion, erosion, and thread or groove depth. See the definition of c in para. K304.1.1(b).

K302.5 Mechanical Strength

Paragraph 302.5 applies.

PART 2
PRESSURE DESIGN OF PIPING COMPONENTS

K303 GENERAL

Components manufactured in accordance with standards listed in Table K326.1.1-1 shall be considered suitable for use at pressure-temperature ratings in accordance with para. K302.2.

K304 PRESSURE DESIGN OF HIGH PRESSURE COMPONENTS

K304.1 Straight Pipe

(24) K304.1.1 General

(a) The required wall thickness of straight sections of pipe shall be determined in accordance with eq. (33).

$$t_m = t + c \quad (33)$$

The minimum wall thickness, T , for the pipe selected, considering manufacturer's minus tolerance, shall be not less than t_m .

(b) The following nomenclature is used in the equation for pressure design of straight pipe:

$c = c_i + c_o$
 = the sum of mechanical allowances² (thread or groove depth) plus corrosion and erosion allowances (where c_i = the sum of *internal* allowances and c_o = the sum of *external* allowances). For threaded components, the nominal thread depth (dimension h of ASME B1.20.1 or equivalent) shall apply, except that for straight threaded connections, the external thread groove depth need not be considered provided all the following criteria are met:

(1) It does not exceed 20% of the wall thickness.

(2) The ratio of outside to inside diameter, D/d , is greater than 1.1.

(3) The internally threaded attachment provides adequate reinforcement.

(4) The thread plus the undercut area, if any, does not extend beyond the reinforcement for a distance more than the nominal wall thickness of the pipe.

t = pressure design wall thickness, as calculated in para. K304.1.2 for internal pressure, or in accordance with the procedure listed in para. K304.1.3 for external pressure

t_m = minimum required wall thickness, including mechanical, corrosion, and erosion allowances

Adequate reinforcement by the attachment is defined as that necessary to ensure that the static burst pressure of the connection will equal or exceed that of the unthreaded portion of the pipe. The adequacy of the reinforcement shall be substantiated as required by para. K304.7.2.

K304.1.2 Straight Pipe Under Internal Pressure. The internal pressure design wall thickness, t , shall be not less than that calculated in accordance with eq. (34a) for pipe with a specified outside diameter and minimum wall

thickness, or eq. (34b) for pipe with a specified inside diameter and minimum wall thickness³

$$t = \frac{D - 2c_o}{2} \left[1 - e^{(-P/S)} \right] \quad (34a)$$

or

$$t = \frac{d + 2c_i}{2} \left[e^{(P/S)} - 1 \right] \quad (34b)$$

Alternatively, the internal design gage pressure, P , may be calculated by eq. (35a) or eq. (35b)³

$$P = S \times \ln \left[\frac{D - 2c_o}{D - 2(T - c_i)} \right] \quad (35a)$$

or

$$P = S \times \ln \left[\frac{d + 2(T - c_o)}{d + 2c_i} \right] \quad (35b)$$

where

D = outside diameter of pipe. For design calculations in accordance with this Chapter, the outside diameter of the pipe is the maximum value allowable under the specifications.

d = inside diameter of pipe. For design calculations in accordance with this Chapter, the inside diameter of the pipe is the maximum value allowable under the specifications.

P = internal design gage pressure

S = allowable stress from Table K-1 (Table B-1C)

T = pipe wall thickness (measured or minimum in accordance with the purchase specification)

K304.1.3 Straight Pipe Under External Pressure. The pressure design thickness for straight pipe under external pressure shall be determined in accordance with para. K304.1.2. Straight pipe under external pressure shall also meet the criteria against buckling given in ASME BPVC, Section VIII, Division 3, Article KD-2, KD-222.

K304.2 Curved and Mitered Segments of Pipe

K304.2.1 Pipe Bends. The minimum required wall thickness, t_m , of a bend, after bending, may be determined as for straight pipe in accordance with para. K304.1, provided that the bend radius of the pipe centerline is equal to or greater than ten times the nominal pipe outside diameter and the tolerances and strain limits of para. K332 are met. Otherwise the design shall be qualified as required by para. K304.7.2.

² For machined surfaces or grooves where the tolerance is not specified, the tolerance shall be assumed to be 0.5 mm (0.02 in.) in addition to the specified depth of the cut.

³ Any mechanical, corrosion, or erosion allowance, c , not specified as internal, c_i , or external, c_o , shall be assumed to be internal, i.e., $c = c_i$ and $c_o = 0$.

K304.2.2 Elbows. Manufactured elbows not in accordance with [para. K303](#) and pipe bends not in accordance with [para. K304.2.1](#) shall be qualified as required by [para. K304.7.2](#).

K304.2.3 Miter Bends. Miter bends are not permitted.

K304.2.4 Curved Segments of Pipe Under External Pressure. The wall thickness of curved segments of pipe subjected to external pressure may be determined as specified for straight pipe in [para. K304.1.3](#).

K304.3 Branch Connections

K304.3.1 General. Acceptable branch connections include a fitting in accordance with [para. K303](#), an extruded outlet in accordance with [para. 304.3.4](#), or a branch connection fitting (see [para. 300.2](#)) similar to that shown in [Figure K328.5.4-1](#).

K304.3.2 Strength of Branch Connections

(a) The opening made for a branch connection reduces both static and fatigue strength of the run pipe. There shall be sufficient material in the branch connection to contain pressure and meet reinforcement requirements.

(b) Static pressure design of a branch connection not in accordance with [para. K303](#) shall conform to [para. 304.3.4](#) for an extruded outlet or shall be qualified as required by [para. K304.7.2](#).

K304.3.3 Reinforcement of Welded Branch Connections. Branch connections made as provided in [para. 304.3.3](#) are not permitted.

K304.4 Closures

(a) Closures not in accordance with [para. K303](#) or (b) below shall be qualified as required by [para. K304.7.2](#).

(b) Closures may be designed in accordance with the methods, allowable stresses, and temperature limits of ASME BPVC, Section VIII, Division 2 or Division 3, and ASME BPVC, Section II, Part D.

K304.5 Pressure Design of Flanges and Blanks

(24) K304.5.1 Flanges — General

(a) Flanges not in accordance with [para. K303](#) or (b) below shall be qualified as required by [para. K304.7.2](#).

(b) A flange may be designed in accordance with the methods, allowable stresses, and temperature limits of ASME BPVC, Section VIII, Division 2, Part 4, 4.16, or Part 5, or Division 3, Article KD-6, and ASME BPVC, Section II, Part D.

(24) K304.5.2 Blind Flanges

(a) Blind flanges not in accordance with [para. K303](#) or (b) or (c) below shall be qualified as required by [para. K304.7.2](#).

(b) A blind flange may be designed in accordance with [eq. \(36\)](#). The thickness of the flange selected shall be not less than t_m (see [para. K304.1.1](#) for nomenclature), considering manufacturing tolerance

$$t_m = t + c \quad (36)$$

The methods, allowable stresses, and temperature limits of ASME BPVC, Section VIII, Division 2, Part 4, 4.6 may be used, with the following changes in nomenclature, to calculate t_m :

c = sum of mechanical allowances, defined in [para. K304.1.1](#)

t = pressure design thickness as calculated for the given style of blind flange using the appropriate equation of ASME BPVC, Section VIII, Division 2, Part 4, 4.6

(c) A blind flange may be designed in accordance with the rules, allowable stresses, and temperature limits of ASME BPVC, Section VIII, Division 3, Article KD-6 and ASME BPVC, Section II, Part D.

K304.5.3 Blanks. Design of blanks shall be in accordance with [para. 304.5.3\(b\)](#), except that E shall be 1.00 and the definitions of S and c shall be in accordance with [para. K304.1.1](#).

K304.6 Reducers

Reducers not in accordance with [para. K303](#) shall be qualified as required by [para. K304.7.2](#).

K304.7 Pressure Design of Other Components

K304.7.1 Listed Components. Other pressure-containing components manufactured in accordance with standards in [Table K326.1.1-1](#) may be utilized in accordance with [para. K303](#).

K304.7.2 Unlisted Components. Pressure design of unlisted components to which the rules elsewhere in [para. K304](#) do not apply shall be based on the pressure design criteria of this Chapter. The designer shall ensure that the pressure design has been substantiated through one or more of the means stated in (a), (b), and (c) below. Note that designs are also required to be checked for adequacy of mechanical strength as described in [para. K302.5](#).

(a) extensive, successful service experience under comparable design conditions with similarly proportioned components made of the same or like material.

(b) performance testing sufficient to substantiate both the static pressure design and fatigue life at the intended operating conditions. Static pressure design may be substantiated by demonstrating that failure or excessive plastic deformation does not occur at a pressure equivalent to two times the internal design pressure, P . The test pressure shall be two times the design pressure multiplied

by the ratio of allowable stress at test temperature to the allowable stress at design temperature, and by the ratio of actual yield strength to the specified minimum yield strength at room temperature from Table K-1 (Table K-1C).

(c) detailed stress analysis (e.g., finite element method) with results evaluated as described in ASME BPVC, Section VIII, Division 3, Article KD-2, except that for linear elastic analysis, the Division 3 stress intensity limits due to combined sustained and occasional loads may be increased by the same factor applied in para. K302.3.6(a).

(d) for (a), (b), and (c) above, interpolations supported by analysis are permitted between sizes, wall thicknesses, and pressure classes, as well as analogies among related materials with supporting material property data. Extrapolation is not permitted.

K304.7.3 Components With Nonmetallic Parts.

Gaskets, packing, seals, and valve seats that meet the applicable requirements of para. A302.1 may be used. Other nonmetallic parts are not permitted.

K304.7.4 Expansion Joints. Expansion joints are not permitted.

K304.8 Fatigue Analysis

K304.8.1 General. A fatigue analysis shall be performed on each piping system, including all components⁴ and joints therein, and considering the stresses resulting from attachments, to determine its suitability for the cyclic operating conditions⁵ specified in the engineering design. Except as permitted in (a) and (b) below, or in paras. K304.8.4 and K304.8.5, this analysis shall be in accordance with ASME BPVC, Section VIII, Division 3.⁶ The cyclic conditions shall include pressure variations as well as thermal variations or displacement stresses. The requirements of para. K304.8 are in addition to the requirements for a flexibility analysis stated in para. K319. No formal fatigue analysis is required in systems that

(a) are duplicates of successfully operating installations or replacements without significant change of systems with a satisfactory service record or

(b) can readily be judged adequate by comparison with previously analyzed systems

K304.8.2 Amplitude of Alternating Stress

(a) The values of the alternating stress intensity, the associated mean stress, and the equivalent alternating stress intensity shall be determined in accordance with ASME BPVC, Section VIII, Division 3, Articles KD-2 and KD-3. The allowable amplitude of the equivalent alter-

nating stress shall be determined from the applicable design fatigue curve in Article KD-3.

(b) If it can be shown that the piping component will fail in a leak-before-burst mode, the number of design cycles (design fatigue life) may be calculated in accordance with either ASME BPVC, Section VIII, Division 3, Article KD-3 or Article KD-4. If a leak-before-burst mode of failure cannot be shown, the fracture mechanics evaluation outlined in Article KD-4 shall be used to determine the number of design cycles of the component.

(c) *Additional Considerations.* The designer is cautioned that the considerations listed in para. K302.1 may reduce the fatigue life of the component below the value predicted by (a) or (b) above.

K304.8.3 Pressure Stress Evaluation for Fatigue Analysis

(a) For fatigue analysis of straight pipe, eq. (37) may be used to calculate the stress intensity⁷ at the inside surface due only to internal pressure

$$S = \frac{PD^2}{2(T - c)[D - (T - c)]} \quad (37)$$

(b) For fatigue analysis of curved pipe, eq. (37) may be used, with the dimensions of the straight pipe from which it was formed, to calculate the maximum stress intensity at the inside surface due only to internal pressure, provided that the centerline bend radius is not less than ten times the nominal outside diameter of the pipe, and that the tolerance and strain limits of para. K332 are met. Bends of smaller radius shall be qualified as required by para. K304.7.2.

(c) If the value of S calculated by eq. (37) exceeds two times S_{yt} at the average temperature during the loading cycle, an inelastic analysis shall be performed. See para. K302.3.2(b) for the definition of S_{yt} .

K304.8.4 Fatigue Evaluation by Test. With the owner's approval, the design fatigue life of a component may be established by destructive testing in accordance with para. K304.7.2 in lieu of the above analysis requirements.

K304.8.5 Extended Fatigue Life. The design fatigue life of piping components may be extended beyond that determined by ASME BPVC, Section VIII, Division 3, Article KD-3 by the use of one of the following methods, provided that the component is qualified in accordance with para. K304.7.2:

(a) surface treatments, such as improved surface finish

(b) prestressing methods, such as autofrettage, shot peening, or shrink fit

⁴ Bore imperfections may reduce fatigue life.

⁵ If the range of temperature change varies, equivalent full temperature cycles N may be computed using eq. (1d) in para. 302.3.5.

⁶ Fatigue analysis in accordance with ASME BPVC, Section VIII, Division 3 requires that stress concentration factors be used in computing the cyclic stresses.

⁷ The term "stress intensity" is defined in ASME BPVC, Section VIII, Division 3.

(24)

Table K305.1.2-1
Required Ultrasonic or Eddy Current Examination of Pipe and Tubing for Longitudinal Defects

Diameter, mm (in.)	Examination Required	Paragraph Reference
$d < 3.2 \left(\frac{1}{8}\right)$ or $D < 6.4 \left(\frac{1}{4}\right)$	None	...
$3.2 \left(\frac{1}{8}\right) \leq d \leq 17.5 \left(\frac{11}{16}\right)$ and $6.4 \left(\frac{1}{4}\right) \leq D \leq 25.4 (1.00)$	Eddy current (ET) [Note (1)] or ultrasonic (UT)	K344.8 or K344.6
$d > 17.5 \left(\frac{11}{16}\right)$ or $D > 25.4 (1.00)$	Ultrasonic (UT)	K344.6

NOTE: (1) This examination is limited to cold drawn austenitic stainless steel pipe and tubing.

The designer is cautioned that the benefits of prestress may be reduced due to thermal, strain softening, or other effects.

PART 3 FLUID SERVICE REQUIREMENTS FOR PIPING COMPONENTS

K305 PIPE

Pipe includes components designated as tube or tubing in the material specification, when intended for pressure service.

K305.1 Requirements

K305.1.1 General. Pipe and tubing shall be either seamless or longitudinally welded with straight seam and a joint quality factor $E_j = 1.00$, examined in accordance with Table K341.3.2-1, Note (2). Spiral (helical seam) welds are not permitted.

K305.1.2 Additional Examination. Pipe and tubing shall have passed a 100% examination for longitudinal defects in accordance with Table K305.1.2-1. This examination is in addition to acceptance tests required by the material specification.

K305.1.3 Heat Treatment. Heat treatment, if required, shall be in accordance with para. K331.

K305.1.4 Unlisted Pipe and Tubing. Unlisted pipe and tubing may be used only in accordance with para. K302.2.3.

K306 FITTINGS, BENDS, AND BRANCH CONNECTIONS

Pipe and other materials used in fittings, bends, and branch connections shall be suitable for the manufacturing or fabrication process and otherwise suitable for the service.

K306.1 Pipe Fittings

K306.1.1 General. All castings shall have examination and acceptance criteria in accordance with para. K302.3.3. All welds shall have a weld quality factor $E_j = 1.00$, with examination and acceptance criteria in accordance with paras. K341 through K344. Spiral (helical seam) welds are not permitted. Listed fittings may be used in accordance with para. K303. Unlisted fittings may be used only in accordance with para. K302.2.3.

K306.1.2 Specific Fittings

- (a) Socket welding fittings are not permitted.
- (b) Threaded fittings are permitted only in accordance with para. K314.
- (c) Branch connection fittings (see para. 300.2) whose design has been performance tested successfully as required by para. K304.7.2(b) may be used within their established ratings.

K306.2 Pipe Bends

K306.2.1 General. A bend made in accordance with para. K332.2 and verified for pressure design in accordance with para. K304.2.1 shall be suitable for the same service as the pipe from which it is made.

K306.2.2 Corrugated and Other Bends. Bends of other design (such as creased or corrugated) are not permitted.

K306.3 Miter Bends

Miter bends are not permitted.

K306.4 Fabricated or Flared Laps

Only forged laps are permitted.

K306.5 Fabricated Branch Connections

Fabricated branch connections constructed by welding shall be fabricated in accordance with para. K328.5.4 and examined in accordance with para. K341.4.

K307 VALVES AND SPECIALTY COMPONENTS

The following requirements for valves shall also be met, as applicable, by other pressure-containing piping components, e.g., traps, strainers, and separators. See also Appendix F, paras. F301.4 and F307.

K307.1 General

K307.1.1 Listed Valves. A listed valve is suitable for use in High Pressure Fluid Service, except as stated in [para. K307.2](#).

K307.1.2 Unlisted Valves. Unlisted valves may be used only in accordance with [para. K302.2.3](#).

K307.2 Specific Requirements

K307.2.1 Bonnet Bolting. The bonnets of bolted bonnet valves shall be secured to their bodies by at least four bolts.

K307.2.2 Stem Retention. [Paragraph 307.2.2](#) applies.

K307.2.3 Double Seated Valves. [Paragraph 307.2.3](#) applies, except for [para. 307.2.3\(b\)](#). Valve cavity design shall be substantiated by one or more of the means stated in [paras. K304.7.2\(b\)](#) and [K304.7.2\(c\)](#) for the increased pressure-temperature condition.

K308 FLANGES, BLANKS, FLANGE FACINGS, AND GASKETS**K308.1 General**

Pressure design of unlisted flanges shall be verified in accordance with [para. K304.5.1](#) or qualified as required by [para. K304.7.2](#).

K308.2 Specific Flanges

K308.2.1 Threaded Flanges. Threaded flanges may be used only within the limitations on threaded joints in [para. K314](#).

K308.2.2 Other Flange Types. Slip-on, socket welding, and expanded joint flanges, and flanges for flared laps, are not permitted.

K308.3 Flange Facings

The flange facing shall be suitable for the service and for the gasket and bolting employed.

K308.4 Gaskets

Gaskets shall be selected so that the required seating load is compatible with the flange rating and facing, the strength of the flange, and its bolting. Materials shall be suitable for the service conditions. Mode of gasket failure shall be considered in gasket selection and joint design.

K308.5 Blanks

Blanks shall have a marking, identifying material, pressure-temperature rating, and size, that is visible after installation.

K309 BOLTING

(24)

Bolting, including bolts, bolt studs, studs, cap screws, nuts, and washers, shall meet the requirements of ASME BPVC, Section VIII, Division 2, Part 3, 3.7; Part 4, 4.16; and Part 5, 5.7. See also [Appendix F, para. F309](#), of this Code.

PART 4**FLUID SERVICE REQUIREMENTS FOR PIPING JOINTS****K310 GENERAL**

Joints shall be suitable for the fluid handled, and for the pressure-temperature and other mechanical loadings expected in service.

Factors such as assembly and disassembly (if applicable), cyclic loading, vibration, shock, bending, and thermal expansion and contraction of joints shall be considered in the engineering design.

K311 WELDED JOINTS**K311.1 General**

Welds shall conform to the following:

- (a) Welding shall be in accordance with [para. K328](#).
- (b) Preheating and heat treatment shall be in accordance with [paras. K330](#) and [K331](#), respectively.
- (c) Examination shall be in accordance with [para. K341.4](#), with acceptance criteria as shown in [Table K341.3.2-1](#).

K311.2 Specific Requirements

K311.2.1 Backing Rings and Consumable Inserts. Backing rings shall not be used. Consumable inserts shall not be used in butt welded joints except when specified by the engineering design.

K311.2.2 Fillet Welds. Fillet welds may be used only for structural attachments in accordance with the requirements of [paras. K321](#) and [K328.5.2](#).

K311.2.3 Other Weld Types. Socket welds and seal welds are not permitted.

K312 FLANGED JOINTS

Flanged joints shall be selected for leak tightness, considering the requirements of [para. K308](#), flange facing finish, and method of attachment. See also [para. F312](#).

K312.1 Joints Using Flanges of Different Ratings

[Paragraph 312.1](#) applies.

K313 EXPANDED JOINTS

Expanded joints are not permitted.

K314 THREADED PIPE JOINTS

For the purposes of this paragraph, “pipe” does not include tube or tubing.

K314.1 General

Except as provided in [paras. K314.2](#) and [K314.3](#), threaded pipe joints are not permitted.

(a) Layout of piping shall be such as to minimize strain on threaded joints that could adversely affect sealing.

(b) Supports shall be designed to control or minimize strain and vibration on threaded joints and seals.

(24) K314.2 Taper-Threaded Pipe Joints

(a) Taper-threaded pipe joints shall be used only for instrumentation, vents, drains, and similar purposes, and shall be not larger than DN 15 (NPS $\frac{1}{2}$).

(b) The nominal wall thickness of piping components with external taper threads shall be at least as thick as that specified for Schedule 160 in ASME B36.10.

K314.3 Straight-Threaded Pipe Joints

K314.3.1 Joints With Seal Formed by Projecting Pipe.

Threaded joints where the threads are used to attach flanges or fittings, and in which the pipe end projects through the flange or fitting and is machined to form the sealing surface with a lens ring, cone ring, or other similar sealing device, may be used. Such joints shall be qualified in accordance with [para. K304.7.2](#).

K314.3.2 Other Straight-Threaded Joints

(a) *Other Joints Using Components Conforming to Listed Standards.* Pipe joints may incorporate straight-threaded fittings conforming to listed standards in [Table K326.1.1-1](#), provided the fittings

(1) are compatible with the pipe with which they are used, considering tolerances and other characteristics

(2) comply with [para. K302.2.1](#) or [para. K302.2.2](#)

(b) *Other Joints Using Components Not Conforming to Listed Standards.* Other straight-threaded pipe joints (e.g., a union comprising external and internal ends joined with a threaded union nut, or other designs shown typically in [Figure 335.3.3-1](#)) may be used. Such joints shall be qualified by performance testing in accordance with [para. K304.7.2\(b\)](#). Testing shall be conducted for each material type/grade and heat treatment condition, component configuration (e.g., elbow), size (e.g., NPS), and pressure rating. Performance testing of joints in which the process of making up the joint involves significant uncontrolled loads (e.g., hammer unions) shall include testing designed to simulate actual loads.

K315 TUBING JOINTS

K315.1 Flared End Tubing Joints

Flared end tubing joints, whether the flare provides the seal, carries the load, or both, may be used, provided the type of fitting selected is adequate for the design pressure, other loadings, and the design temperature. The design shall also be qualified in accordance with [para. K304.7.2](#).

K315.2 Flareless Tubing Joints Using Components Conforming to Listed Standards

Tubing joints may incorporate flareless type fittings conforming to listed standards in [Table K326.1.1-1](#), provided the fittings

(a) are compatible with the tubing with which they are used, considering tolerances and other characteristics

(b) comply with [para. K302.2.1](#) or [para. K302.2.2](#)

K315.3 Flareless Tubing Joints Using Components Not Conforming to Listed Standards

Tubing joints may incorporate flareless type fittings not conforming to listed standards in [Table K326.1.1-1](#), provided the type of fitting selected is adequate for the design pressure, other loadings, and the design temperature, and meets the requirements of [para. K302.2.3](#).

K316 CAULKED JOINTS

Caulked joints are not permitted.

K317 SOLDERED AND BRAZED JOINTS

K317.1 Soldered Joints

Soldered joints are not permitted.

K317.2 Brazed and Braze Welded Joints

(a) Braze welded joints and fillet joints made with brazing filler metal are not permitted.

(b) Brazed joints shall be made in accordance with [para. K333](#) and shall be qualified as required by [para. K304.7.2](#). Such application is the owner’s responsibility. The melting point of brazing alloys shall be considered when exposure to fire is possible.

K318 SPECIAL JOINTS

Special joints are those not covered elsewhere in this Part.

K318.1 General

Joints may be used in accordance with [para. 318.2](#) and the requirements for materials and components in this Chapter.

K318.2 Specific Requirements

K318.2.1 Prototype Tests. A prototype joint shall have been subjected to performance tests in accordance with [para. K304.7.2\(b\)](#) to determine the safety of the joint under test conditions simulating all expected service conditions. Testing shall include cyclic simulation.

K318.2.2 Prohibited Joints. Bell type and adhesive joints are not permitted.

PART 5 FLEXIBILITY AND SUPPORT

K319 FLEXIBILITY

Flexibility analysis shall be performed for each piping system. [Paragraphs 319.1](#) through [319.6](#) apply, except for [para. 319.4.1\(c\)](#). The computed displacement stress range shall be within the allowable displacement stress range in [para. K302.3.5](#) and shall also be included in the fatigue analysis in accordance with [para. K304.8](#).

K320 ANALYSIS OF SUSTAINED LOADS

K320.1 Basic Assumptions and Requirements

[Paragraph 320.1](#) applies, but refer to [para. K302.3.5\(c\)](#) instead of [para. 302.3.5\(c\)](#).

K320.2 Stress Due to Sustained Loads

[Paragraph 320.2](#) applies, except that references to expansion joints are not applicable.

K321 PIPING SUPPORT

Piping supports and methods of attachment shall be in accordance with [para. 321](#) except as modified below, and shall be detailed in the engineering design.

K321.1 General

K321.1.1 Objectives. [Paragraph 321.1.1](#) applies, but substitute “Chapter” for “Code” in [321.1.1\(a\)](#).

K321.1.4 Materials. [Paragraph 321.1.4](#) applies, but replace [321.1.4\(e\)](#) with the following:

(e) Attachments welded to the piping shall be of a material compatible with the piping and the service. Other requirements are specified in [paras. K321.3.2](#) and [K323.4.2\(b\)](#).

K321.3 Structural Attachments

K321.3.2 Integral Attachments. [Paragraph 321.3.2](#) applies, but substitute “K321.1.4(e)” for “321.1.4(e)” and “Chapter IX” for “Chapter V.”

PART 6 SYSTEMS

K322 SPECIFIC PIPING SYSTEMS

K322.3 Instrument Piping

K322.3.1 Definition. Instrument piping within the scope of this Chapter includes all piping and piping components used to connect instruments to high pressure piping or equipment. Instruments, permanently sealed fluid-filled tubing systems furnished with instruments as temperature- or pressure-responsive devices, and control piping for air or hydraulically operated control apparatus (not connected directly to the high pressure piping or equipment) are not within the scope of this Chapter.

K322.3.2 Requirements. Instrument piping within the scope of this Chapter shall be in accordance with [para. 322.3.2](#) except that the design pressure and temperature shall be determined in accordance with [para. K301](#), and the requirements of [para. K310](#) shall apply. Instruments, and control piping not within the scope of this Chapter, shall be designed in accordance with [para. 322.3](#).

K322.6 Pressure-Relieving Systems

[Paragraph 322.6](#) applies, except for [para. 322.6.3](#).

K322.6.3 Overpressure Protection. Overpressure protection for high pressure piping systems shall conform to the following:

(a) The cumulative capacity of pressure relief devices shall be sufficient to prevent the pressure from rising more than 10% above the design pressure at the operating temperature during the relieving condition.

(b) System protection shall include one pressure relief device with a set pressure at or below the design pressure at the operating temperature for the relieving condition. The set pressure of additional pressure relief devices shall not exceed 105% of the design pressure.

PART 7 MATERIALS

K323 GENERAL REQUIREMENTS

(a) [Paragraph K323](#) states limitations and required qualifications for materials based on their inherent properties. Their use is also subject to requirements elsewhere in [Chapter IX](#) and in [Table K-1 \(Table K-1C\)](#).

(b) Specific attention should be given to the manufacturing process to ensure uniformity of properties throughout each piping component.

(c) See [para. K321.1.4](#) for support materials.

K323.1 Materials and Specifications

K323.1.1 Listed Materials

(a) Any material used in a pressure-containing piping component shall conform to a listed specification, except as provided in (b) below or in para. K323.1.2.

(b) Materials manufactured to specification editions different from those listed in Appendix E may be used, provided

(1) the requirements for chemical composition and heat-treatment condition in the edition of the specification to which the material was manufactured meet the requirements of the listed edition

(2) the specified minimum tensile and yield strengths, and, if applicable, the specified maximum tensile and yield strengths, required by the two editions of the specification are the same, and

(3) the material has been tested and examined in accordance with the requirements of the listed edition of the specification

A material that does not meet the requirements of (b)(1), (b)(2), and (b)(3) may be evaluated as an unlisted material in accordance with para. K323.1.2.

K323.1.2 Unlisted Materials. An unlisted material may be used, provided it conforms to a published specification covering chemistry, physical and mechanical properties, method and process of manufacture, heat treatment, and quality control, and otherwise meets the requirements of this Chapter. Allowable stresses shall be determined in accordance with the applicable allowable stress basis of this Chapter or a more conservative basis.

K323.1.3 Unknown Materials. Materials of unknown specification, type, or grade are not permitted.

K323.1.4 Reclaimed Materials. Reclaimed pipe and other piping components may be used provided they are properly identified as conforming to a listed specification, have documented service history for the material and fatigue life evaluation, and otherwise meet the requirements of this Chapter. Sufficient cleaning and inspection shall be made to determine minimum wall thickness and freedom from defects that would be unacceptable in the intended service.

K323.1.5 Product Analysis. Conformance of materials to the product analysis chemical requirements of the applicable specification shall be verified, and certification shall be supplied. Requirements for product analysis are defined in the applicable materials specification.

K323.1.6 Repair of Materials by Welding. A material defect may be repaired by welding, provided that all of the following criteria are met:

- (a) The material specification provides for weld repair.
- (b) The welding procedure and welders or welding operators are qualified as required by para. K328.2.

(c) The repair and its examination are performed in accordance with the material specification and with the owner's approval.

K323.2 Temperature Limitations

The designer shall verify that materials that meet other requirements of this Chapter are suitable for service throughout the operating temperature range.

K323.2.1 Upper Temperature Limits, Listed Materials. A listed material shall not be used at a temperature above that shown in the Max. Temp. column of Table K-1 (Table K-1C). (24)

K323.2.2 Lower Temperature Limits, Listed Materials

(a) The design minimum temperature for listed materials and their welds shall not be colder than the impact test temperature determined in accordance with para. K323.3.4(a), except as provided in (b) or (c) below.

(b) For listed materials and their welds subjected to neither longitudinal nor circumferential stresses greater than 41 MPa (6 ksi), the design minimum temperature shall not be colder than the lower of -48°C (-55°F) and the impact test temperature determined in accordance with para. K323.3.4(a).

(c) For listed materials and their welds exempted from Charpy impact testing by para. K323.3.1(a), the design minimum temperature shall not be colder than -198°C (-325°F) for fully austenitic stainless steels, or -48°C (-55°F) for other materials.

K323.2.3 Temperature Limits, Unlisted Materials. An unlisted material acceptable under para. K323.1.2 shall be qualified for service at all temperatures within a stated range from design minimum temperature to design (maximum) temperature, in accordance with para. K323.2.4. However, the upper temperature limit shall be less than the temperature for which an allowable stress, determined in accordance with para. 302.3.2, is governed by the creep or stress rupture provisions of that paragraph.

K323.2.4 Verification of Serviceability

(a) When an unlisted material is used, the designer is responsible for demonstrating the validity of the allowable stresses and other design limits, and of the approach taken in using the material, including the derivation of stress data and the establishment of temperature limits.

(b) Paragraph 323.2.4(b) applies except that allowable stress values shall be determined in accordance with para. K302.3.

K323.3 Impact Testing Methods and Acceptance Criteria

(24) K323.3.1 General

(a) Metallic piping components and their welds used in High Pressure Fluid Service shall be subjected to Charpy V-notch impact testing as described in (b), with the following exceptions:

- (1) nuts
- (2) washers
- (3) gaskets, including lens rings, cone rings, and other similar sealing devices

(4) piping components meeting the requirements of a standard listed in Table K326.1.1-1 that includes pressure-temperature ratings, and the design conditions are within these ratings. For such components, the impact test methods and acceptance criteria of para. 323.3 apply.

(5) other piping components and their welds where the maximum obtainable longitudinal Charpy V-notch specimen (see para. K323.3.3) has a width along the notch less than 2.5 mm (0.098 in.)

(b) Charpy impact testing shall be performed in accordance with Table K323.3.1-1 on representative samples using the testing methods described in paras. K323.3.2 through K323.3.4. Acceptance criteria are described in para. K323.3.5.

K323.3.2 Procedure. Paragraph 323.3.2 applies.

K323.3.3 Test Specimens

(a) Each set of impact test specimens shall consist of three specimen bars. Impact tests shall be made using standard 10 mm (0.394 in.) square cross section Charpy V-notch specimen bars oriented in the transverse direction.

(b) Where component size, shape, or both do not permit specimens as specified in (a) above, standard 10 mm square cross-section longitudinal Charpy specimens may be prepared.

(c) Where component size, shape, or both do not permit specimens as specified in (a) or (b) above, subsize longitudinal Charpy specimens may be prepared. Test temperature shall be reduced in accordance with Table 323.3.4-1.

(d) If necessary in (a), (b), or (c) above, corners of specimens parallel to and on the side opposite the notch may be as shown in Figure K323.3.3-1.

K323.3.4 Test Temperatures. For all Charpy impact tests, the test temperature criteria in (a) or (c) below shall be observed.

(a) Charpy impact tests shall be conducted at a temperature no warmer than the coldest of the following:

- (1) 21°C (70°F)
- (2) when the design minimum temperature is warmer than or equal to -48°C (-55°F), the lowest component temperature expected in service at which a material

or weld will be subjected to a longitudinal or circumferential stress greater than 41 MPa (6 ksi)

(3) when the design minimum temperature is colder than -48°C (-55°F), the design minimum temperature

(b) In specifying the temperatures in (a)(2) and (a)(3), the following shall be considered:

- (1) range of operating conditions
- (2) upset conditions
- (3) ambient temperature extremes
- (4) required leak test temperature

(c) Where the largest possible test specimen has a width along the notch less than the lesser of 80% of the material thickness or 8 mm (0.315 in.), the test shall be conducted at a reduced temperature in accordance with Table 323.3.4-1, considering the temperature as reduced below the test temperature required by (a).

K323.3.5 Acceptance Criteria

(a) *Minimum Energy Requirements for Materials Other Than Bolting.* The applicable minimum impact energy requirements for materials shall be those shown in Table K323.3.5-1. Lateral expansion shall be measured in accordance with ASTM A370 (for title see para. 323.3.2). The results shall be included in the impact test report.

(b) *Minimum Energy Requirements for Bolting Materials.* The applicable minimum energy requirements shall be those shown in Table K323.3.5-1 except as provided in Table K323.3.1-1.

(c) *Weld Impact Test Requirements.* Where two base metals having different required impact energy values are joined by welding, the impact test energy requirements shall equal or exceed the requirements of the base material having the lower required impact energy.

(d) Retests

(1) *Retest for Absorbed Energy Criteria.* When the average value of the three specimens equals or exceeds the minimum value permitted for a single specimen, and the value for more than one specimen is below the required average value, or when the value for one specimen is below the minimum value permitted for a single specimen, a retest of three additional specimens shall be made. The value for each of these retest specimens shall equal or exceed the required average value.

(2) *Retest for Erratic Test Results.* When an erratic result is caused by a defective specimen or uncertainty in the test, a retest will be allowed. The report giving test results shall specifically state why the original specimen was considered defective or which step of the test procedure was carried out incorrectly.

(24)

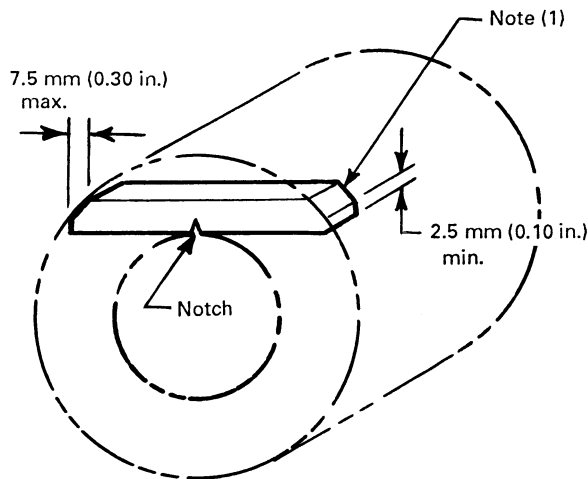
Table K323.3.1-1
Impact Testing Requirements

Test Characteristics		Column A Pipe, Tubes, and Components Made From Pipe or Tubes	Column B Other Components, Fittings, Etc.	Column C Bolts
Tests on Materials	Number of tests	As required by the material specification, or one test set per lot [Note (1)], whichever is greater, except as permitted by para. K323.3.1(a).		
	Location and orientation of specimens [Note (2)]	(a) Transverse to the longitudinal axis, with notch parallel to axis. [Note (3)] (b) Where component size, shape, or both do not permit specimens as specified in (a) above, paras. K323.3.3(b) through K323.3.3(d) apply as needed.	(a) Transverse to the direction of maximum elongation during rolling or to direction of major working during forging. Notch shall be oriented parallel to direction of maximum elongation or major working. [Note (3)] (b) If there is no single identifiable axis, e.g., for castings or triaxial forgings, specimens shall either meet the longitudinal values of Table K323.3.5-1, or three sets of orthogonal specimens shall be prepared, and the lowest impact values obtained from any set shall meet the transverse values of Table K323.3.5-1. (c) Where component size, shape, or both do not permit specimens as specified in (a) or (b) above, paras. K323.3.3(c) and K323.3.3(d) apply as needed.	(a) Bolts ≤52 mm (≤2 in.) nominal size made in accordance with ASTM A320 shall meet the impact requirements of that specification. (b) For all other bolts, longitudinal specimens shall be taken. The impact values obtained shall meet the transverse values of Table K323.3.5-1.
Tests on Welds in Fabrication or Assembly [Note (4)]	Procedure qualification thickness limits	Impact tests shall be valid only if the thickness of the test material meets the requirements of ASME BPVC, Section IX, Table QW-451.1 (including Notes), except that if the thickness is less than 16 mm ($\frac{5}{8}$ in.), the thickness of the test material shall be the minimum thickness qualified.		
	Testing requirements [Note (5)]	Impact tests shall be performed on the weld metal and the HAZ of the procedure qualification test coupon at a test temperature no warmer than that determined in accordance with para. K323.3.4.		
	Location and orientation of specimens	(1) Weld metal impact specimens shall be taken across the weld with the notch in the weld metal. Each specimen shall be oriented so that the notch axis is normal to the surface of the material and one face of the specimen shall be within 1.5 mm ($\frac{1}{16}$ in.) of the surface of the material. (2) Heat-affected zone impact specimens shall be taken across the weld and have sufficient length to locate the notch in the heat-affected zone, after etching. The notch shall be cut approximately normal to the material surface in such a manner as to include as much heat-affected zone material as possible in the resulting fracture.		
	Acceptance criteria	(1) Impact tests shall meet the acceptance criteria applicable to the base metal in accordance with para. K323.3.5. (2) The impact values obtained from both the weld metal and HAZ specimens shall meet the transverse values in Table K323.3.5-1.		

NOTES:

- (1) A lot shall consist of pipe or components of the same nominal size, made from the same heat of material, and heat treated together. If a continuous type furnace is used, pipe or components may be considered to have been heat treated together if they are processed during a single continuous time period at the same furnace conditions.
- (2) Impact tests shall be performed on a representative sample of material after completion of all heat treatment and forming operations involving plastic deformation, except that cold bends made in accordance with para. K304.2.1 need not be tested after bending.
- (3) For welded piping components, specimens shall be taken from the base metal, weld metal, and the heat-affected zone.
- (4) For welds in the fabrication or assembly of piping or components, including repair welds.
- (5) The test piece shall be large enough to permit preparing the number of specimens required by para. K323.3. If this is not possible, additional test pieces shall be prepared.

Figure K323.3.3-1
Example of an Acceptable Impact Test Specimen



GENERAL NOTE: This Figure illustrates how an acceptable transverse Charpy specimen can be obtained from a tubing or component shape too small for a full length standard specimen in accordance with ASTM A370. The corners of a longitudinal specimen parallel to and on the side opposite the notch may be as shown.

NOTE: (1) Corners of the Charpy specimen [see [para. K323.3.3\(d\)](#)] may follow the contour of the component within the dimension limits shown.

K323.4 Requirements for Materials

K323.4.1 General. Requirements in [para. K323.4](#) apply to pressure-containing parts, not to materials used as supports, gaskets, packing, or bolting. See also [Appendix F, para. F323.4](#).

K323.4.2 Specific Requirements

- (a) Ductile iron and other cast irons are not permitted.
- (b) Zinc-coated materials are not permitted for pressure containing components and may not be attached to pressure-containing components by welding.

(24) **K323.4.3 Metallic Clad and Lined Materials.** Materials with metallic cladding or lining may be used in accordance with the following provisions:

- (a) For metallic clad or lined piping components, the base metal shall be an acceptable material as defined in [para. K323](#), and the thickness used in pressure design in accordance with [para. K304](#) shall not include the thickness of the cladding or lining. The allowable stress used shall be that for the base metal at the design temperature. For such components, the cladding or lining may be any material that, in the judgment of the user, is suitable for the intended service and for the method of manufacture and assembly of the piping component.

(b) Fabrication by welding of clad or lined piping components and the inspection and testing of such components shall be done in accordance with applicable provisions of ASME BPVC, Section VIII, Division 1, Part UCL, UCL-30 through UCL-52, and the provisions of this Chapter.

(c) If a metallic liner also serves as a gasket or as part of the flange facing, the requirements and limitations in [para. K308.4](#) apply.

K323.5 Deterioration of Materials in Service

[Paragraph 323.5](#) applies.

K325 MISCELLANEOUS MATERIALS

[Paragraph 325](#) applies.

PART 8 STANDARDS FOR PIPING COMPONENTS

K326 REQUIREMENTS FOR COMPONENTS

K326.1 Dimensional Requirements

K326.1.1 Listed Piping Components. Dimensional standards for piping components are listed in [Table K326.1.1-1](#). Dimensional requirements contained in specifications listed in [Appendix K](#) shall also be considered requirements of this Chapter.

K326.1.2 Unlisted Piping Components. Piping components not listed in [Table K326.1.1-1](#) or [Appendix K](#) shall meet the pressure design requirements described in [para. K302.2.3](#) and the mechanical strength requirements described in [para. K302.5](#).

K326.1.3 Threads. The dimensions of piping connection threads not otherwise covered by a governing component standard or specification shall conform to the requirements of applicable standards listed in [Table K326.1.1-1](#) or [Appendix K](#).

K326.2 Ratings of Components

K326.2.1 Listed Components. The pressure-temperature ratings of components listed in [Table K326.1.1-1](#) are accepted for pressure design in accordance with [para. K303](#).

K326.2.2 Unlisted Components. The pressure-temperature ratings of unlisted piping components shall conform to the applicable provisions of [para. K304](#).

K326.3 Reference Documents

The documents listed in [Table K326.1.1-1](#) contain references to codes, standards, and specifications not listed in [Table K326.1.1-1](#). Such unlisted codes, standards, and

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Table K323.3.5-1
Minimum Required Charpy V-Notch Impact Values

Specimen Orientation	Pipe Wall or Component Thickness, mm (in.)	Number of Specimens [Note (1)]	Energy, J (ft-lbf), for Specified Minimum Yield Strength, MPa (ksi) [Note (2)]	
			≤932 (≤135)	>932 (>135)
Transverse	≤25 (≤1.0)	Average for 3	27 (20)	34 (25)
		Minimum for 1	20 (15)	27 (20)
	>25 and ≤51 (>1.0 and ≤2.0)	Average for 3	34 (25)	41 (30)
		Minimum for 1	27 (20)	33 (24)
	>51 (>2.0)	Average for 3	41 (30)	47 (35)
		Minimum for 1	33 (24)	38 (28)
Longitudinal	≤25 (≤1.0)	Average for 3	54 (40)	68 (50)
		Minimum for 1	41 (30)	54 (40)
	>25 and ≤51 (>1.0 and ≤2.0)	Average for 3	68 (50)	81 (60)
		Minimum for 1	54 (40)	65 (48)
	>51 (>2.0)	Average for 3	81 (60)	95 (70)
		Minimum for 1	65 (48)	76 (56)

NOTES:

(1) See [para. K323.3.5\(c\)](#) for permissible retests.

(2) Energy values in this table are for standard size specimens. For subsize specimens, these values shall be multiplied by the ratio of the actual specimen width to that of a full-size specimen, 10 mm (0.394 in.).

specifications shall be used only in the context of the listed documents in which they appear.

The design, materials, fabrication, assembly, examination, inspection, and testing requirements of this Chapter are not applicable to components manufactured in accordance with the documents listed in [Table K326.1.1-1](#), unless specifically stated in this Chapter or in the listed document.

K326.4 Repair of Piping Components by Welding

A defect in a piping component may be repaired by welding subject to all of the following requirements:

(a) The piping component specification provides for weld repair or, if not covered by a specification, the manufacturer allows for weld repair.

(b) The welding procedure and welders or welding operators are qualified as required by [para. K328.2](#).

(c) The repair and its examination are performed in accordance with the piping component specification or, if not covered by a specification, the manufacturer's requirements.

(d) The owner approves the weld repair.

PART 9 FABRICATION, ASSEMBLY, AND ERECTION

K327 GENERAL

Piping materials and components are prepared for assembly and erection by one or more of the fabrication processes covered in [paras. K328, K330, K331, K332, and K333](#). When any of these processes is used in assembly or erection, requirements are the same as for fabrication.

K328 WELDING

Welding shall conform to the requirements of this Part and the applicable requirements of [para. K311](#).

K328.1 Welding Responsibility

Each employer is responsible for

(a) the welding performed by personnel of its organization

(b) conducting the qualification tests required to qualify the welding procedure specifications used by personnel in its organization

(c) conducting the qualification tests required to qualify its welders and welding operators

Table K326.1.1-1
Component Standards

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Designator	Title
Component Standards Having Established Ratings	
Fittings, Flanges, Gaskets, Line Blanks, and Valves	
ASME B16.5	Pipe Flanges and Flanged Fittings: NPS 1/2 Through NPS 24 Metric/Inch Standard [Notes (1), (2)]
ASME B16.20	Metallic Gaskets for Pipe Flanges
ASME B16.34	Valves — Flanged, Threaded, and Welding End [Notes (1), (2)]
ASME B16.48	Line Blanks [Notes (1), (2)]
MSS SP-65	High-Pressure Chemical Industry Flanges and Threaded Stubs for Use with Lens Gaskets [Notes (1), (2)]
Component Standards Not Having Specific Ratings	
Fittings	
ASME B16.9	Factory-Made Wrought Butt welding Fittings [Notes (3), (4)]
ASME B16.11	Forged Fittings, Socket-Welding and Threaded [Notes (3), (4)]
Dimensional Standards	
Bolting	
ASME B18.2.1	Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)
ASME B18.2.2	Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)
Pipe	
ASME B36.10	Welded and Seamless Wrought Steel Pipe
ASME B36.19	Welded and Seamless Wrought Stainless Steel Pipe
Threads	
API 5B	Threading, Gauging, and Inspection of Casing, Tubing, and Line Pipe Threads
ASME B1.1	Unified Inch Screw Threads (UN, UNR, and UNJ Thread Forms)
ASME B1.20.1	Pipe Threads, General Purpose (Inch)
Welding End Preparation for Piping Components	
ASME B16.25	Butt welding Ends

GENERAL NOTE: The approved edition dates of these standards and specifications, along with the names and websites of the sponsoring organizations, are shown in [Appendix E](#).

NOTES:

- (1) Use of these standards does not require calculations in accordance with this Chapter. Where materials not listed in [Appendix K](#) are to be used, they need not be qualified as unlisted materials in accordance with [para. K323.1.2](#). However, materials shall not exceed the upper temperature limit in [para. K323.2.1](#) or [para. K323.2.3](#).
- (2) Piping components fully complying with these standards, including the pressure-temperature ratings contained therein, may be used, provided
 - (a) they meet the fluid service requirements in [Parts 3](#) and [4](#) of this Chapter.
 - (b) they are examined and leak tested in accordance with the requirements of [paras. K341](#) and [K345](#), respectively.
- (3) Use of these standards requires calculations in accordance with this Chapter. Where materials not listed in [Appendix K](#) are to be used, they shall be qualified as unlisted materials in accordance with [para. K323.1.2](#).
- (4) Piping components made in accordance with these standards may be used, provided
 - (a) their pressure design is in accordance with the requirements of [para. K302.2.2\(a\)](#).
 - (b) they meet the fluid service requirements in [Parts 3](#) and [4](#) of this Chapter.
 - (c) they are examined and leak tested in accordance with the requirements of [paras. K341](#) and [K345](#), respectively.
 - (d) they are Charpy V-notch impact tested in accordance with the methods described in [paras. K323.3.1](#) through [K323.3.4](#) and meet the acceptance criteria specified in [para. K323.3.5](#). Note that such impact testing may require the destruction of one component from the same lot as the component to be used in service [see [Table K323.3.1-1](#), Note (1)].
 - (e) the upper temperature limit in [para. K323.2.1](#) or [para. K323.2.3](#) is not exceeded.

K328.2 Welding Qualifications

K328.2.1 Qualification Requirements. Qualification of the welding procedures to be used and of the performance of welders and welding operators shall comply with the requirements of ASME BPVC, Section IX, except as modified herein.

(a) Impact tests shall be performed for all procedure qualifications in accordance with [para. K323.3](#).

(b) Except as provided in (f), test weldments shall be made using the same specification and type or grade of base metal(s), and the same specification and classification of filler metal(s) as will be used in production welding.

(c) Test weldments shall be subjected to essentially the same heat treatment, including cooling rate and cumulative time at temperature, as the production welds.

(d) When tensile specimens are required by ASME BPVC, Section IX, the yield strength shall also be determined, using the method required for the base metal. The yield strength of each test specimen shall be not less than the specified minimum yield strength at room temperature (S_Y) for the base metals joined. Where two base metals having different S_Y values are joined by welding, the yield strength of each test specimen shall be not less than the lower of the two S_Y values.

(e) Mechanical testing is required for all performance qualification tests.

(f) If the same specification and classification of filler metal(s) as will be used in production welding are employed, qualification on pipe or tubing shall also qualify for plate of the same type or grade, but qualification on plate does not qualify for pipe or tubing.

K328.2.2 Procedure Qualification by Others. Qualification of welding procedures by others is not permitted.

K328.2.3 Performance Qualification by Others. Welding performance qualification by others is not permitted.

K328.2.4 Qualification Records. [Paragraph 328.2.4](#) applies.

K328.3 Materials

(24) **K328.3.1 Electrodes and Filler Metal.** [Paragraph 328.3.1](#) applies, together with the following additional requirements:

(a) Welding electrodes and filler metals shall be specified in the engineering design.

(b) With the owner's approval, a welding electrode or filler metal not conforming to a specification listed in ASME BPVC, Section II, Part C may be used if a procedure qualification test, including an all-weld-metal test in accordance with AWS B4.0M or AWS B4.0⁸, is first successfully made.

⁸ AWS B4.0M or AWS B4.0. Standard Methods for Mechanical Testing of Welds. American Welding Society.

K328.3.2 Weld Backing Material. Backing rings shall not be used.

K328.3.3 Consumable Inserts. [Paragraph 328.3.3](#) applies, except that procedures shall be qualified as required by [para. K328.2](#).

K328.4 Preparation for Welding

K328.4.1 Cleaning. [Paragraph 328.4.1](#) applies.

K328.4.2 End Preparation

(a) *General*

(1) Butt weld end preparation is acceptable only if the surface is machined or ground to bright metal.

(2) Butt welding end preparation contained in ASME B16.25 or any other end preparation that meets the procedure qualification is acceptable. [For convenience, the basic bevel angles taken from ASME B16.25, with some additional J-bevel angles, are shown in [Figure 328.4.2-1](#), illustrations (a) and (b).]

(b) *Circumferential Welds*

(1) If components' ends are trimmed as shown in [Figure 328.4.2-1](#), illustration (a) or illustration (b) to accommodate consumable inserts, or as shown in [Figure K328.4.3-1](#) to correct internal misalignment, such trimming shall not result in a finished wall thickness before welding less than the required minimum wall thickness, t_m .

(2) It is permissible to size pipe ends of the same nominal size to improve alignment, if wall thickness requirements are maintained.

(3) Where necessary, weld metal may be deposited on the inside or outside of the component to permit alignment or provide for machining to ensure satisfactory seating of inserts.

(4) When a butt weld joins sections of unequal wall thickness and the thicker wall is more than $1\frac{1}{2}$ times the thickness of the other, end preparation and geometry shall be in accordance with acceptable designs for unequal wall thickness in ASME B16.5.

K328.4.3 Alignment

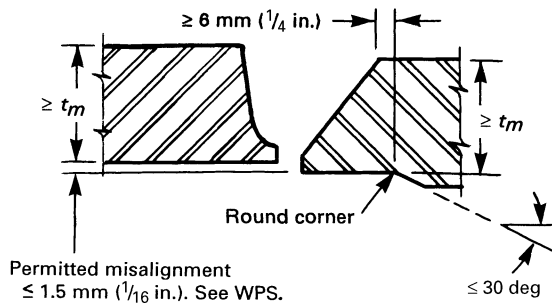
(a) *Girth Butt Welds*

(1) Inside diameters of components at the ends to be joined shall be aligned within the dimensional limits in the welding procedure and the engineering design, except that no more than 1.5 mm ($\frac{1}{16}$ in.) misalignment is permitted as shown in [Figure K328.4.3-1](#).

(2) If the external surfaces of the two components are not aligned, the weld shall be tapered between the two surfaces with a slope not steeper than 1:4.

(b) *Longitudinal Butt Joints.* Preparation for longitudinal butt welds (not made in accordance with a standard listed in [Table K-1](#) ([Table K-1C](#)) or [Table K326.1.1-1](#)) shall conform to the requirements of (a).

Figure K328.4.3-1
Pipe Bored for Alignment:
Trimming and Permitted Misalignment



(c) Branch Connection Welds

(1) The dimension m in Figure K328.5.4-1 shall not exceed ± 1.5 mm ($1/16$ in.).

(2) The dimension g in Figure K328.5.4-1 shall be specified in the engineering design and the welding procedure.

K328.5 Welding Requirements

K328.5.1 General. The requirements of paras. 328.5.1(b), 328.5.1(d), 328.5.1(e), and 328.5.1(f) apply in addition to the requirements specified below.

(a) All welds, including tack welds, repair welds, and the addition of weld metal for alignment [paras. K328.4.2(b)(3) and K328.4.3(c)(1)], shall be made by qualified welders or welding operators, in accordance with a qualified procedure.

(b) Tack welds at the root of the joint shall be made with filler metal equivalent to that used for the root pass. Tack welds shall be fused with the root pass weld, except that those that have cracked shall be removed. Bridge tacks (above the root) shall be removed.

K328.5.2 Fillet Welds. Fillet welds, where permitted (see para. K311.2.2), shall be fused with and shall merge smoothly into the component surfaces.

K328.5.3 Seal Welds. Seal welds are not permitted.

K328.5.4 Welded Branch Connections. Branch connection fittings (see para. 300.2), attached by smoothly contoured full penetration groove welds of a design that permits 100% interpretable radiographic examination, are the only types acceptable.

Figure K328.5.4-1 shows acceptable details of welded branch connections. The illustrations are typical and are not intended to exclude acceptable details not shown.

K328.5.5 Fabricated Laps. Fabricated laps are not permitted.

K328.6 Weld Repair

Paragraph 328.6 applies, except that procedures and performance shall be qualified as required by para. K328.2.1. See also para. K341.3.3.

K328.7 Attachment Welds

(24)

Paragraph 328.7 applies, except refer to paras. K341.4.1(d) and K341.3.2 instead of paras. 341.4 and 341.3.2, respectively.

K330 PREHEATING

K330.1 General

The preheat requirements herein apply to all types of welding, including tack welds and repair welds.

K330.1.1 Requirements. Paragraph 330.1.1 applies.

K330.1.2 Unlisted Materials. Paragraph 330.1.2 applies.

K330.1.3 Preheat Temperature Verification. Paragraph 330.1.3(a) applies. Temperature-indicating materials and techniques shall not be detrimental to the base metals. (24)

K330.1.4 Preheat Zone. Paragraph 330.1.4 applies.

K330.2 Specific Requirements

Paragraph 330.2 applies in its entirety.

K331 HEAT TREATMENT

K331.1 General

K331.1.1 Postweld Heat Treatment Requirements. The provisions of para. 331.1.1 and Tables 331.1.1-1 through 331.1.3-1 apply, except as specified below.

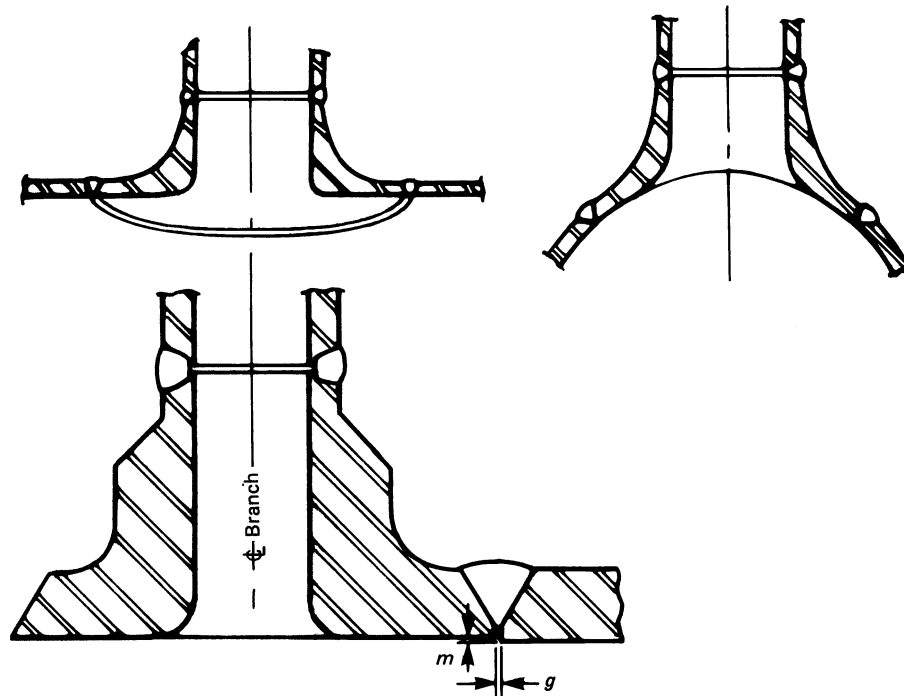
(a) The exemptions to mandatory postweld heat treatment in Table 331.1.3-1 for P-No. 1, all Group Nos., are not permitted when the control thickness exceeds 19 mm ($3/4$ in.).

(b) The exemptions to mandatory postweld heat treatment in Table 331.1.3-1 for P-No. 4, Group No. 1 and P-No. 5A, Group No. 1 materials are not permitted.

(c) For quenched and tempered materials, any heat treatment of other than longitudinal welds shall be performed at a temperature no warmer than 28°C (50°F) below the tempering temperature of the material. When this heat treatment temperature is lower than the range specified in Table 331.1.1-1, the decrease in heat treatment temperature shall be accompanied by an increase in holding time in accordance with Table 331.1.2-1.

(d) Longitudinal welds in quenched and tempered material shall be heat treated in accordance with the applicable material specification.

Figure K328.5.4-1
Some Acceptable Welded Branch Connections Suitable for 100% Radiography



- (e) Refer to [para. K331.1.3](#) instead of [para. 331.1.3](#).
- (f) Refer to [Appendix K](#) instead of [Appendix A](#).
- (g) ASME B31P and [para. 328.7](#) do not apply.

K331.1.2 Other Heat Treatments. Heat treatment for bending and forming shall be in accordance with [para. K332.4](#).

K331.1.3 Definition of Thicknesses Governing Post-weld Heat Treatment. The provisions of [para. 331.1.3](#) apply, except for [para. 331.1.3\(b\)\(5\)](#).

K331.1.4 Heating and Cooling. [Paragraph 331.1.4](#) applies.

K331.1.6 Temperature Verification. Heat treatment temperature shall be checked by thermocouple pyrometers or other suitable methods to ensure that the WPS requirements are met. Temperature-indicating materials and techniques shall not be detrimental to the base metals.

K331.2 Specific Requirements

[Paragraph 331.2](#) applies, but refer to [para. K331.1](#) instead of [para 331.1](#).

K332 BENDING AND FORMING

K332.1 General

Pipe shall be hot or cold bent in accordance with a written procedure to any radius that will result in surfaces free of cracks and free of buckles. The procedure shall address at least the following, as applicable:

- (a) material specification and range of size and thickness
- (b) range of bend radii and fiber elongation
- (c) minimum and maximum metal temperature during bending
- (d) method of heating and maximum hold time
- (e) description of bending apparatus and procedure to be used
- (f) mandrels or material and procedure used to fill the bore
- (g) method for protection of thread and machined surfaces
- (h) examination to be performed
- (i) required heat treatment
- (j) postheat treatment dimensional adjustment technique

K332.2 Bending

K332.2.1 Bend Ovality. [Paragraph 332.2.1](#) applies. (24)

K332.2.2 Bending Temperature. Paragraph 332.2.2 applies, except that in cold bending of quenched and tempered ferritic materials, the temperature shall be at least 28°C (50°F) below the tempering temperature.

K332.3 Forming

Piping components shall be formed in accordance with a written procedure. The temperature range shall be consistent with material characteristics, end use, and specified heat treatment. The thickness after forming shall be not less than required by design. The procedure shall address at least the following, as applicable:

- (a) material specification and range of size and thickness
- (b) maximum fiber elongation expected during forming
- (c) minimum and maximum metal temperature during bending
- (d) method of heating and maximum hold time
- (e) description of forming apparatus and procedure to be used
- (f) materials and procedures used to provide internal support during forming
- (g) examination to be performed
- (h) required heat treatment

K332.4 Required Heat Treatment

K332.4.1 Hot Bending and Forming. After hot bending and forming, heat treatment is required for all thicknesses of P-Nos. 3, 4, 5A, 5B, 6, 10A, 10B, and 15E materials that are not quenched and tempered. Times and temperatures shall be in accordance with para. 331. Quenched and tempered materials shall be reheat treated to the original material specification.

K332.4.2 Cold Bending and Forming

(a) After cold bending and forming, heat treatment in accordance with (b) below is required, regardless of thickness, when specified in the engineering design or when the maximum calculated fiber elongation exceeds 5% strain or 50% of the basic minimum specified longitudinal elongation for the applicable specification, grade, and thickness for P-Nos. 1, 3, 4, 5A, 5B, 6, 10A, 10B, and 15E materials (unless it has been demonstrated that the selection of the pipe and the procedure for making the components provide assurance that the most severely formed portion of the material has retained an elongation of not less than 10%).

(b) Heat treatment is required regardless of thickness and shall conform to the temperatures and durations given in Table 331.1.1-1, except that for quenched and tempered materials, the stress relieving temperature shall not exceed a temperature 28°C (50°F) below the tempering temperature of the material.

K333 BRAZING AND SOLDERING

Brazing shall conform to the requirements of this Part and the applicable requirements of para. K317.2. Braze welding and soldering are not permitted.

K333.1 Brazing Responsibility

Each employer is responsible for

- (a) the brazing performed by personnel of its organization
- (b) conducting the qualification tests required to qualify the brazing procedure specifications used by personnel in its organization
- (c) conducting the qualification tests required to qualify its brazers and brazing operators

K333.2 Brazing Qualifications

K333.2.1 Qualification Requirements. The qualification of brazing procedures, brazers, and brazing operators shall be in accordance with the requirements of ASME BPVC, Section IX.

K333.2.2 Procedure Qualification by Others. Brazing procedure specifications qualified by a technically competent group or agency may be used, provided the following requirements are met:

- (a) The owner shall approve the use of brazing procedure specifications qualified by others.
- (b) The brazing procedure specifications shall meet the requirements of para. K333.2.1.
- (c) The employer's business name shall be shown on each brazing procedure specification and on each brazing procedure qualification record.
- (d) The employer shall accept responsibility for each brazing procedure qualification performed by others by signing and dating each brazing procedure qualification record.
- (e) The employer shall conduct the qualification tests required to qualify its brazers and brazing operators.

K333.2.3 Performance Qualification by Others. An employer may accept the performance qualification of a brazer or brazing operator made by a technically competent group or agency, provided the following requirements are met:

- (a) The owner shall approve the use of brazing performance qualifications qualified by others.
- (b) The brazer or brazing operator performance qualifications were made on pipe test coupons.
- (c) The employer shall have a brazing procedure specification such that the performance qualification is within the limits of the essential variables set forth in ASME BPVC, Section IX.
- (d) The employer shall have a copy of the performance qualification test record. The record shall show the name of the technically competent group or agency by whom the brazer or brazing operator was qualified and the date of

that qualification. Evidence shall also be provided that the brazer or brazing operator has maintained qualification in accordance with ASME BPVC, Section IX, Article XIII, QB-322.

(e) The employer's business name shall be shown on the brazing performance qualification record.

(f) The employer shall accept responsibility for the brazing performance qualifications performed by others by signing and dating the qualification.

K333.2.4 Qualification Records. The employer shall maintain copies of the brazing procedure specification and performance qualification records specified by ASME BPVC, Section IX. These records shall be made available to the Inspector upon request.

K333.3 Materials

Brazing filler metals and flux shall be in accordance with [para. 333.2.1](#).

K333.4 Preparation and Cleaning

[Paragraph 333.3](#) applies, except that soldering is not permitted.

(24) K335 ASSEMBLY AND ERECTION

[Paragraphs 335.1, 335.2, 335.9, and 335.10](#) apply.

K335.3 Threaded Joints

[Paragraph 335.3](#) applies, except that threaded joints shall not be seal welded.

K335.6 Special Joints

Special joints (as defined in [para. K318](#)) shall be installed and assembled in accordance with the manufacturer's instructions, as modified by the engineering design. Care shall be taken to ensure full engagement of joint members.

PART 10 INSPECTION, EXAMINATION, AND TESTING

K340 INSPECTION

[Paragraphs 340.1 through 340.4](#) apply.

K341 EXAMINATION

[Paragraphs 341.1 and 341.2](#) apply.

K341.3 Examination Requirements

K341.3.1 General. Prior to initial operation, each piping installation, including components and workmanship, shall be examined in accordance with [para. K341.4](#) and the engineering design. If heat treatment is

performed, examination shall be conducted after its completion.

K341.3.2 Acceptance Criteria. Acceptance criteria (24) shall be as stated in the engineering design and shall at least meet the applicable requirements stated in (a) through (c) below, and elsewhere in this Chapter.

(a) [Table K341.3.2-1](#) states acceptance criteria (limits on imperfections) for welds. See [Figure 341.3.2-1](#) for typical weld imperfections.

(b) Acceptance criteria for castings are specified in [para. K302.3.3](#).

(c) Acceptance criteria for surfaces on pressure-containing components where attachment welds have been removed are specified in [para. 341.3.2\(c\)](#).

K341.3.3 Defective Components and Workmanship.

Defects (imperfections of a type or magnitude not acceptable by the criteria specified in [para. K341.3.2](#)) shall be repaired, or the defective item or work shall be replaced.

Examination shall be as follows:

(a) When the defective item or work is repaired, the repaired portion of the item or work shall be examined after the completion of any required heat treatment. The examination shall use the same methods and acceptance criteria employed for the original examination.

(b) When the defective item or work is replaced, the new item or work used to replace the defective item or work shall be examined after the completion of any required heat treatment. The examination shall use any method and applicable acceptance criteria that meet the requirements for the original examination.

K341.4 Extent of Required Examination

(24)

Piping shall be examined to the extent specified herein or to any greater extent specified in the engineering design.

K341.4.1 Visual Examination

(a) The following shall be 100% examined in accordance with [para. 344.2](#):

(1) all materials and components, to verify that they conform to specifications and are free from defects

(2) all fabrication, as defined in [para. 300.2](#)

(3) all completed welds, except those in components made in accordance with a listed standard

(4) all threaded, bolted, and other mechanical joints, to verify that they conform to the applicable requirements of [para. K335](#)

(5) all piping erection, to verify dimensions and alignment, including that of cold spring in [para. 335.1\(b\)](#) prior to assembly

(6) after assembly, all supports, guides, and points of cold spring, to ensure that movement of the piping under all conditions of startup, operation, and shutdown will be accommodated without undue binding or unanticipated constraint

Table K341.3.2-1
Acceptance Criteria for Welds

(24)

Imperfection	Examination Methods		Criteria (A to F) for Type of Welds [Note (1)]			
	Visual	Ultrasonics or Radiography	Girth Groove	Longitudinal Groove [Note (2)]	Fillet [Note (3)]	Branch Connection [Note (4)]
Crack	✓	✓	A	A	A	A
Incomplete fusion	✓	✓	A	A	A	A
Incomplete penetration	✓	✓	A	A	A	A
Rounded indications	...	✓	B	B	N/A	B
Linear indication	...	✓	C	C	N/A	C
Undercutting	✓	✓	A	A	A	A
Surface porosity or exposed slag inclusion	✓	...	A	A	A	A
Concave surface, concave root surface, or burn-through	✓	✓	D	D	N/A	D
Surface finish	✓	...	E	E	E	E
Face reinforcement or root reinforcement	✓	...	F	F	F	F

GENERAL NOTES:

- (a) Imperfections are evaluated by one or more of the examination methods given, as specified in [paras. K341.4.1](#) and [K341.4.2](#).
 (b) "N/A" indicates this Chapter does not establish acceptance criteria or does not require evaluation of this kind of imperfection for this type of weld.
 (c) Check (✓) indicates examination method generally used for evaluating this kind of imperfection.
 (d) Ellipsis (...) indicates examination method not generally used for evaluating this kind of imperfection.
 (e) Symbols A through F are explained in the table on the next page.

NOTES:

- (1) Criteria given are for required examination. More-stringent criteria may be specified in the engineering design.
 (2) Longitudinal welds include only those permitted in [paras. K302.3.4](#) and [K305](#). The criteria shall be met by all welds, including those made in accordance with a standard listed in [Table K326.1.1-1](#) or in [Appendix K](#).
 (3) Fillet welds include only those permitted in [para. K311.2.2](#).
 (4) Branch connection welds include only those permitted in [para. K328.5.4](#).

Criterion Value Notes for Table K341.3.2-1

Criterion										
Symbol	Measure	Acceptable Value Limits [Note (1)]								
A	Extent of imperfection	Zero (no evident imperfection)								
B	Size and distribution of rounded indications	See ASME BPVC, Section VIII, Division 1, Appendix 4 [Note (2)]								
C	Linear indications characterized as neither incomplete penetration nor incomplete fusion. Linear indications are unacceptable if the amplitude exceeds the reference level, or linear indications have lengths that exceed Individual length [Note (3)] Cumulative length	6 mm ($\frac{1}{4}$ in.) for $\bar{T}_w \leq 19$ mm ($\frac{3}{4}$ in.) $\bar{T}_w/3$ for 19 mm ($\frac{3}{4}$ in.) $< \bar{T}_w \leq 57$ mm ($2\frac{1}{4}$ in.) 19 mm ($\frac{3}{4}$ in.) for $\bar{T}_w > 57$ mm ($2\frac{1}{4}$ in.) $\leq \bar{T}_w$ in any 12 $\bar{T}_w$ weld length [Note (2)]								
D	Depth of surface concavity, root surface concavity, or burn-through	<table><tr><th>$\bar{T}_w$, mm (in.)</th><th>Depth, mm (in.)</th></tr><tr><td>≤ 13 ($\leq \frac{1}{2}$)</td><td>≤ 1.5 ($\leq \frac{1}{16}$)</td></tr><tr><td>> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)</td><td>≤ 3 ($\leq \frac{1}{8}$)</td></tr><tr><td>> 51 (> 2.0)</td><td>≤ 4 ($\leq \frac{5}{32}$)</td></tr></table> and total joint thickness including weld reinforcement $\geq \bar{T}_w$	$\bar{T}_w$, mm (in.)	Depth, mm (in.)	≤ 13 ($\leq \frac{1}{2}$)	≤ 1.5 ($\leq \frac{1}{16}$)	> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)	≤ 3 ($\leq \frac{1}{8}$)	> 51 (> 2.0)	≤ 4 ($\leq \frac{5}{32}$)
$\bar{T}_w$, mm (in.)	Depth, mm (in.)									
≤ 13 ($\leq \frac{1}{2}$)	≤ 1.5 ($\leq \frac{1}{16}$)									
> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)	≤ 3 ($\leq \frac{1}{8}$)									
> 51 (> 2.0)	≤ 4 ($\leq \frac{5}{32}$)									
E	Surface roughness	$\leq 12.5 \mu\text{m}$ ($\leq 500 \mu\text{in.}$) R_a (see ASME B46.1 for definition of roughness average, R_a)								
F	Height of face reinforcement or root reinforcement [Note (4)] in any plane through the weld shall be within the limits of the applicable height value in the tabulation at the right. Weld metal shall be fused with and merge smoothly into the component surfaces.	<table><tr><th>$\bar{T}_w$, mm (in.)</th><th>Height, mm (in.)</th></tr><tr><td>≤ 13 ($\leq \frac{1}{2}$)</td><td>≤ 1.5 ($\leq \frac{1}{16}$)</td></tr><tr><td>> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)</td><td>≤ 3 ($\leq \frac{1}{8}$)</td></tr><tr><td>> 51 (> 2.0)</td><td>≤ 4 ($\leq \frac{5}{32}$)</td></tr></table>	$\bar{T}_w$, mm (in.)	Height, mm (in.)	≤ 13 ($\leq \frac{1}{2}$)	≤ 1.5 ($\leq \frac{1}{16}$)	> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)	≤ 3 ($\leq \frac{1}{8}$)	> 51 (> 2.0)	≤ 4 ($\leq \frac{5}{32}$)
$\bar{T}_w$, mm (in.)	Height, mm (in.)									
≤ 13 ($\leq \frac{1}{2}$)	≤ 1.5 ($\leq \frac{1}{16}$)									
> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)	≤ 3 ($\leq \frac{1}{8}$)									
> 51 (> 2.0)	≤ 4 ($\leq \frac{5}{32}$)									

NOTES:

- (1) Where two limiting values are given, the lesser measured value governs acceptance. $\bar{T}_w$ is the nominal wall thickness of the thinner of two components joined by a butt weld.
- (2) Porosity and inclusions such as slag or tungsten are defined as rounded indications where the maximum length is three times the width or less. These indications may be circular, elliptical, or irregular in shape; may have tails; and may vary in density. Indications where the length is greater than three times the width are defined as linear indications and may also be slag, porosity, or tungsten.
- (3) For ultrasonic examination, refer to para. K344.6.3 for acceptable value limits.
- (4) For groove welds, height is the lesser of the measurements made from the surfaces of the adjacent components. For fillet welds, height is measured from the theoretical throat; root reinforcement does not apply. Required thickness t_m shall not include face reinforcement or root reinforcement.

(7) all erected piping, for evidence of defects that would require repair or replacement, and for other evident deviations from the intent of the design

(b) *Pressure-Containing Threads.* 100% examination for finish and fit is required. Items with visible imperfections in thread finish or any of the following defects shall be rejected:

(1) *Tapered Threads.* Failure to meet gaging requirements in API Spec 5B or ASME B1.20.1, as applicable.

(2) *Straight Threads.* Excessively loose or tight fit when gaged for light interference fit.

(c) *Brazed Joints.* 100% of all brazed joints shall be examined by in-process examination in accordance with para. K344.7.

(d) *Attachment Welds.* Surfaces on pressure-containing components where attachment welds have been removed shall be subjected to visual examination in accordance with para. 344.2.

K341.4.2 Radiographic and Ultrasonic Examination

(a) All girth, longitudinal, and branch connection welds shall be 100% radiographically examined, except as permitted in (b) below.

(b) When specified in the engineering design and with the owner's approval, ultrasonic examination of welds may be substituted for radiographic examination where $\bar{T}_w \geq 13 \text{ mm}$ ($\frac{1}{2} \text{ in.}$).

(c) In-process examination (see para. 344.7) shall not be substituted for radiographic or ultrasonic examination of welds.

K341.4.3 Magnetic Particle and Liquid Penetrant Examination. Surfaces on pressure-containing components where attachment welds have been removed shall be subjected to magnetic particle or liquid penetrant examination in accordance with para. K344.3 or para. K344.4, respectively.

K341.4.4 Certifications and Records. Paragraph 341.4.1(c) applies.

K341.5 Supplementary Examination

Any of the examination methods described in para. K344 may be specified by the engineering design to supplement the examination required by para. K341.4. The extent of supplementary examination to be performed and any acceptance criteria that differ from those specified in para. K341.3.2 shall be specified in the engineering design.

K341.5.1 Hardness Tests. Paragraph 341.5.2 applies.

K341.5.2 Examinations to Resolve Uncertainty. Paragraph 341.5.3 applies.

K342 EXAMINATION PERSONNEL

Paragraph 342 applies in its entirety.

K343 EXAMINATION PROCEDURES

Paragraph 343 applies, except that the examination methods shall comply with para. K344.

K344 EXAMINATION METHODS

K344.1 General

(24)

Paragraph 344.1 applies in its entirety.

K344.2 Visual Examination

Paragraph 344.2 applies in its entirety.

K344.3 Magnetic Particle Examination

The method for magnetic particle examination shall be as specified in

(a) paragraph K302.3.3(b) for castings

(b) ASME BPVC, Section V, Article 7 for welds and other components

K344.4 Liquid Penetrant Examination

The method for liquid penetrant examination shall be as specified in

(a) paragraph K302.3.3(b) for castings

(b) ASME BPVC, Section V, Article 6 for welds and other components

K344.5 Radiographic Examination

The method for radiographic examination shall be as specified in

(a) paragraph K302.3.3(c) for castings

(b) ASME BPVC, Section V, Article 2 for welds and other components

K344.6 Ultrasonic Examination

K344.6.1 Castings. The method for ultrasonic examination of castings shall be as specified in para. K302.3.3(c).

K344.6.2 Pipe and Tubing

(24)

(a) *Method.* Pipe and tubing, required or selected in accordance with Table K305.1.2-1 to undergo ultrasonic examination, shall pass a 100% examination for longitudinal defects in accordance with ASTM E213.⁹ Longitudinal (axial) reference notches shall be introduced on the outer and inner surfaces of the calibration (reference) standard in accordance with Figure 3(c) of ASTM E213 to a depth not greater than the larger of 0.10 mm (0.004 in.) or 4% of specimen thickness and a length not more than 10 times the notch depth.

(b) *Acceptance Criteria.* Any indication greater than that produced by the calibration notch represents a defect; defective pipe and tubing shall be rejected.

⁹ ASTM E213. Standard Practice for Ultrasonic Testing of Metal Pipe and Tubing. ASTM International.

(c) *Records.* For pipe and tubing that passes this examination, a report shall be prepared that contains at least the information specified in 14.2.1 through 14.2.6 of ASTM E213.⁹

(24) K344.6.3 Welds

(a) *Method.* The method for ultrasonic examination of welds shall be as specified in ASME BPVC, Section V, Article 4 for nominal thickness, $\bar{T}_w$, greater than or equal to 13 mm ($\frac{1}{2}$ in.) but less than 25 mm (1.0 in.), and ASME BPVC, Section VIII, Division 3, Article KE-3, KE-301 and KE-302 for nominal wall thickness, $\bar{T}_w$, greater than or equal to 25 mm (1.0 in.).

(b) *Acceptance Criteria.* Cracks, incomplete fusion, incomplete penetration, or undercutting are unacceptable regardless of size or length (see Table K341.3.2-1). In addition, for a linear indication

(1) for nominal wall thickness, $\bar{T}_w$, greater than or equal to 13 mm ($\frac{1}{2}$ in.) but less than 25 mm (1.0 in.), the acceptance criterion for the thickness to be examined specified in para. 344.6.2 applies.

(2) for nominal wall thickness, $\bar{T}_w$, greater than or equal to 25 mm (1.0 in.), the acceptance criteria specified in ASME BPVC, Section VIII, Division 3, Article KE-3, KE-333 for the thickness to be examined apply.

K344.7 In-Process Examination

Paragraph 344.7 applies in its entirety.

(24) K344.8 Eddy Current Examination

K344.8.1 Method. The method for eddy current examination of pipe and tubing shall follow the general guidelines of ASME BPVC, Section V, Article 8, subject to the following specific requirements:

(a) Cold drawn austenitic stainless steel pipe and tubing, selected in accordance with Table K305.1.2-1 for eddy current examination, shall pass a 100% examination for longitudinal defects.

(b) A calibration (reference) standard shall be prepared from a representative sample. A longitudinal (axial) reference notch shall be introduced on the inner surface of the standard to a depth not greater than the larger of 0.10 mm (0.004 in.) or 5% of specimen thickness and a length not more than 6.4 mm (0.25 in.).

K344.8.2 Acceptance Criteria. Any indication greater than that produced by the calibration notch represents a defect; defective pipe or tubing shall be rejected.

K344.8.3 Records. For pipe and tubing that passes this examination, a report shall be prepared that includes at least the following information:

- (a) material identification by type, size, lot, heat, etc.
- (b) listing of examination equipment and accessories
- (c) details of examination technique (including examination speed and frequency) and end effects, if any

(d) description of the calibration standard, including dimensions of the notch, as measured

(e) examination results

K345 LEAK TESTING

K345.1 Required Leak Test

(24)

Prior to initial operation and after completion of the examinations required by para. K341, each piping system shall be leak tested.

(a) Each weld and each piping component, except bolting and individual gaskets to be used during final system assembly and pressure relief devices to be used during operation, shall be hydrostatically or pneumatically leak tested in accordance with para. K345.4 or para. K345.5, respectively. The organization conducting the test shall ensure that during the required leak testing of components and welds, adequate protection is provided to prevent injury to people and damage to property from missile fragments, shock waves, or other consequences of any failure that might occur in the pressurized system.

(b) In addition to the requirements of (a) above, a leak test of the installed piping system, excluding pressure relief devices to be used during operation, shall be conducted at a pressure not less than 110% of the design pressure to ensure tightness, except as provided in (c) or (d) below.

(c) If the leak test required in (a) above is conducted on the installed piping system, the additional test in (b) above is not required.

(d) With the owner's approval, pressure relief devices to be used during operation may be included in the leak test required in (b) above. The leak test pressure may be reduced to prevent the operation of, or damage to, the pressure relief devices, but shall not be less than 90% of the lowest set pressure of the pressure relief devices in the system.

(e) For closure welds, examination in accordance with para. K345.2.3(c) may be substituted for the leak test required in (a) above.

(f) None of the following leak tests may be used in lieu of the leak tests required in para. K345.1:

- (1) initial service leak test (see para. 345.7)
- (2) sensitive leak test (see para. 345.8)
- (3) alternative leak test (see para. 345.9)

K345.2 General Requirements for Leak Tests

Paragraphs 345.2.4 through 345.2.7 apply. See below for paras. K345.2.1, K345.2.2, and K345.2.3.

K345.2.1 Limitations on Pressure

(a) *Through-Thickness Yielding.* If the test pressure would produce stress (exclusive of stress intensification) in excess of S_{yt} at the outside surface of a component at test temperature, as determined by calculation or by testing in

accordance with [para. K304.7.2\(b\)](#), the test pressure may be reduced to the maximum pressure that will result in a stress (exclusive of stress intensification) at the outside surface that will not exceed S_{yt} .

(b) The provisions of [paras. 345.2.1\(b\)](#) and [345.2.1\(c\)](#) apply.

- (24) **K345.2.2 Other Test Requirements.** Paragraph [345.2.2](#) applies. In addition, except for those piping components exempted from impact testing in [para. K323.3.1\(a\)](#), the minimum metal temperature during leak testing shall be not less than the impact test temperature (see [para. K323.3.4](#)).

- (24) **K345.2.3 Special Provisions for Leak Testing.** Paragraphs (a), (b), and (c) below apply only to the leak test specified in [para. K345.1\(a\)](#). They are not applicable to the installed piping system leak test specified in [para. K345.1\(b\)](#).

(a) *Piping Components and Subassemblies.* Piping components and subassemblies may be leak tested either separately or as assembled piping.

(b) *Reassembled Mechanical Joints.* The following mechanical joints need not be subjected to an additional leak test in accordance with [para. K345.1\(a\)](#) if disconnected and reassembled after leak testing:

(1) mechanical joints used to connect piping components and subassemblies that have previously been leak tested

(2) mechanical joints used to connect temporary piping components for the purpose of isolating equipment, instruments, or other piping during the leak test

(c) *Closure Welds.* Leak testing of the final weld connecting piping systems or components that have been successfully leak tested is not required, provided the weld is examined in-process in accordance with [para. 344.7](#) and passes the required 100% radiographic examination in accordance with [para. K341.4.2](#).

(d) See also [Appendix F, para. F345.2.3](#).

K345.3 Preparation for Leak Test

[Paragraph 345.3](#) applies in its entirety.

K345.4 Hydrostatic Leak Test

[Paragraph 345.4.1](#) applies. See [paras. K345.4.2](#) and [K345.4.3](#).

K345.4.2 Test Pressure for Components and Welds. Except as provided in [para. K345.4.3](#), the hydrostatic test pressure at every point in a metallic piping system shall be as follows:

- (a) not less than 1.25 times the design pressure.
 (b) when the design temperature is greater than the test temperature, the minimum test pressure, at the point under consideration, shall be calculated by [eq. \(38\)](#)

$$P_T = 1.25PS_T/S \quad (38)$$

where

P = internal design gage pressure

P_T = minimum test gage pressure

S = allowable stress at component design temperature for the prevalent pipe material; see [Table K-1 \(Table K-1C\)](#)

S_T = allowable stress at test temperature for the prevalent pipe material; see [Table K-1 \(Table K-1C\)](#)

(c) in those cases where the piping system may not include pipe itself, any other component in the piping system, other than pipe-supporting elements and bolting, may be used to determine the S_T/S ratio based on the applicable allowable stresses obtained from [Table K-1 \(Table K-1C\)](#). In those cases where the piping system may be made up of equivalent lengths of more than one material, the S_T/S ratio shall be based on the minimum calculated ratio of the included materials.

K345.4.3 Hydrostatic Test of Piping With Vessels as a System. [Paragraph 345.4.3\(a\)](#) applies.

K345.5 Pneumatic Leak Test

[Paragraph 345.5](#) applies, except for [para. 345.5.4](#). See [para. K345.5.4](#).

K345.5.4 Test Pressure. The pneumatic test pressure for components and welds shall be identical to that required for the hydrostatic test in accordance with [para. K345.4.2](#).

K345.6 Hydrostatic-Pneumatic Leak Test for Components and Welds

If a combination hydrostatic-pneumatic leak test is used, the requirements of [para. K345.5](#) shall be met, and the pressure in the liquid-filled part of the piping shall not exceed the limits stated in [para. K345.4.2](#).

K346 RECORDS

K346.1 Responsibility

It is the responsibility of the designer, the manufacturer, the fabricator, and the erector, as applicable, to prepare the records required by this Chapter and by the engineering design.

K346.2 Required Records

At least the following records, as applicable, shall be provided to the owner or the Inspector by the person responsible for their preparation:

- (a) the engineering design
 (b) the written report required in [para. K300\(b\)\(2\)](#)
 (c) material certifications

(d) procedures used for fabrication, welding, brazing, heat treatment, examination, and testing

(e) repair records of materials and piping components, including the welding procedure used for each, and location of repairs

(f) performance qualifications for welders, brazers, and welding and brazing operators

(g) qualifications of examination personnel

(h) records of examination of pipe and tubing for longitudinal defects as specified in [paras. K344.6.2\(c\)](#) and [K344.8.3](#), as applicable

K346.3 Retention of Records

The owner shall retain one set of the required records for at least 5 yr after they are received.